Creating storage grades, storage pools, EC profiles, and regions

StorageGRID 11.5

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Table of Contents

Creating storage grades, storage pools, EC profiles, and regions . . . 1
Creating and assigning storage grades . . . 1
Configuring storage pools . . . 3
Using Cloud Storage Pools . . . 14
Configuring Erasure Coding profiles . . . 41
Configuring regions (optional and S3 only) . . . 50

Creating storage grades, storage pools, EC profiles, and regions

Before you can create the ILM rules for your StorageGRID system, you must define object storage locations, determine the types of copies you want, and optionally configure S3 regions.

- Creating and assigning storage grades
- Configuring storage pools
- Using Cloud Storage Pools
- Configuring Erasure Coding profiles
- Configuring regions (optional and S3 only)

Creating and assigning storage grades

Storage grades identify the type of storage used by a Storage Node. You can create storage grades if you want ILM rules to place certain objects on certain Storage Nodes, instead of on all nodes at the site. For example, you might want certain objects to be stored on your fastest Storage Nodes, such as StorageGRID all-flash storage appliances.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.

About this task

If you use more than one type of storage, you can optionally create a storage grade to identify each type. Creating storage grades allows you to select a specific type of Storage Node when configuring storage pools.

If storage grade is not a concern (for example, all Storage Nodes are identical), you can skip this procedure and use the All Storage Nodes default storage grade when configuring storage pools.

When you add a new Storage Node in an expansion, that node is added to the All Storage Nodes default storage grade. As a result:

- If an ILM rule uses a storage pool with the All Storage Nodes grade, the new node can be used immediately after the expansion completes.
- If an ILM rule uses a storage pool with a custom storage grade, the new node will not be used until you manually assign the custom storage grade to the node, as described below.

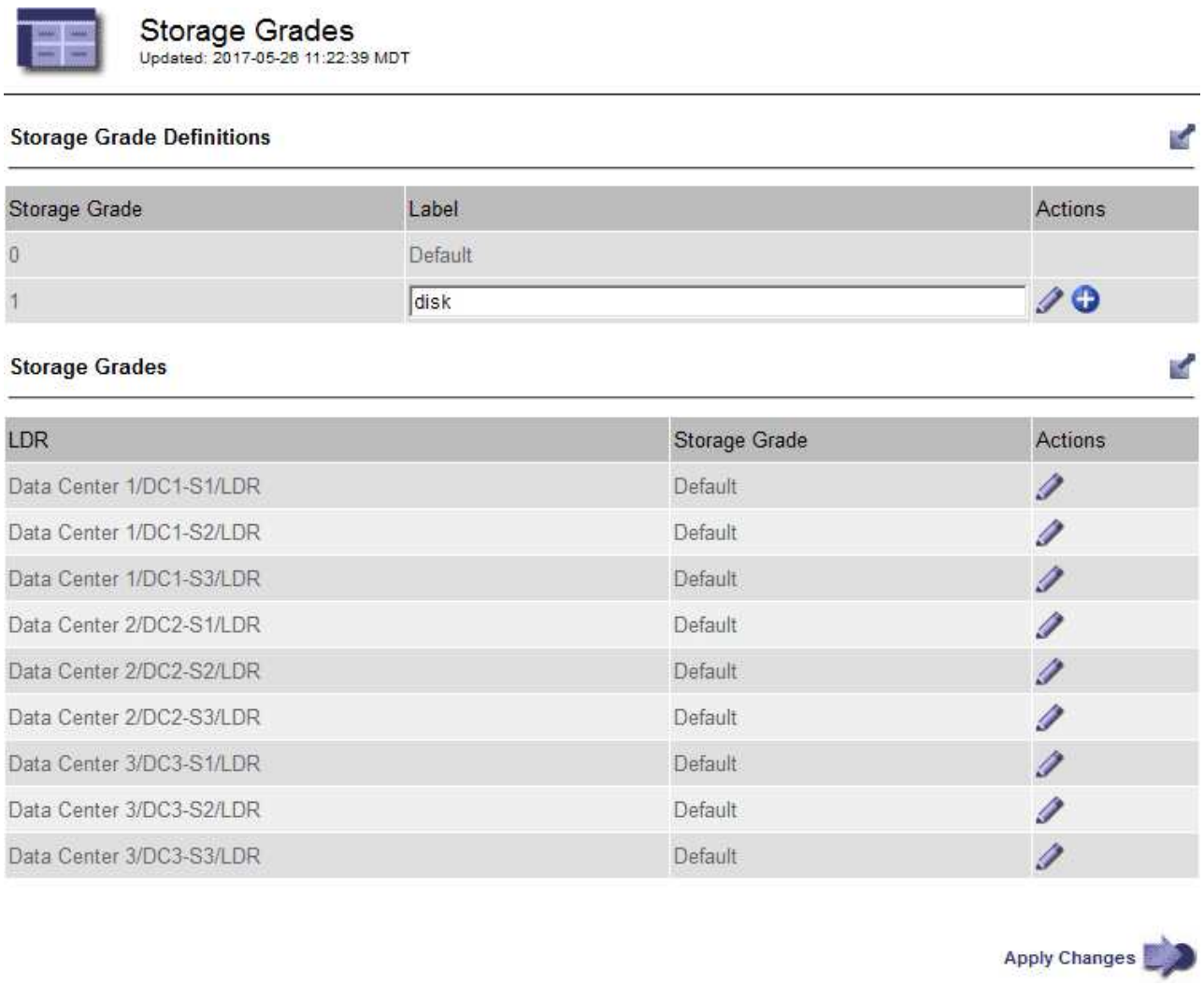
> When creating storage grades, do not create more storage grades than necessary. For example, do not create one storage grade for each Storage Node. Instead, assign each storage grade to two or more nodes. Storage grades assigned to only one node can cause ILM backlogs if that node becomes unavailable.

Steps

1. Select **ILM** > **Storage Grades**.

2. Create a storage grade:
   a. For each storage grade you need to define, click **Insert** to add a row and enter a label for the storage grade.

      The Default storage grade cannot be modified. It is reserved for new Storage Nodes added during a StorageGRID system expansion.



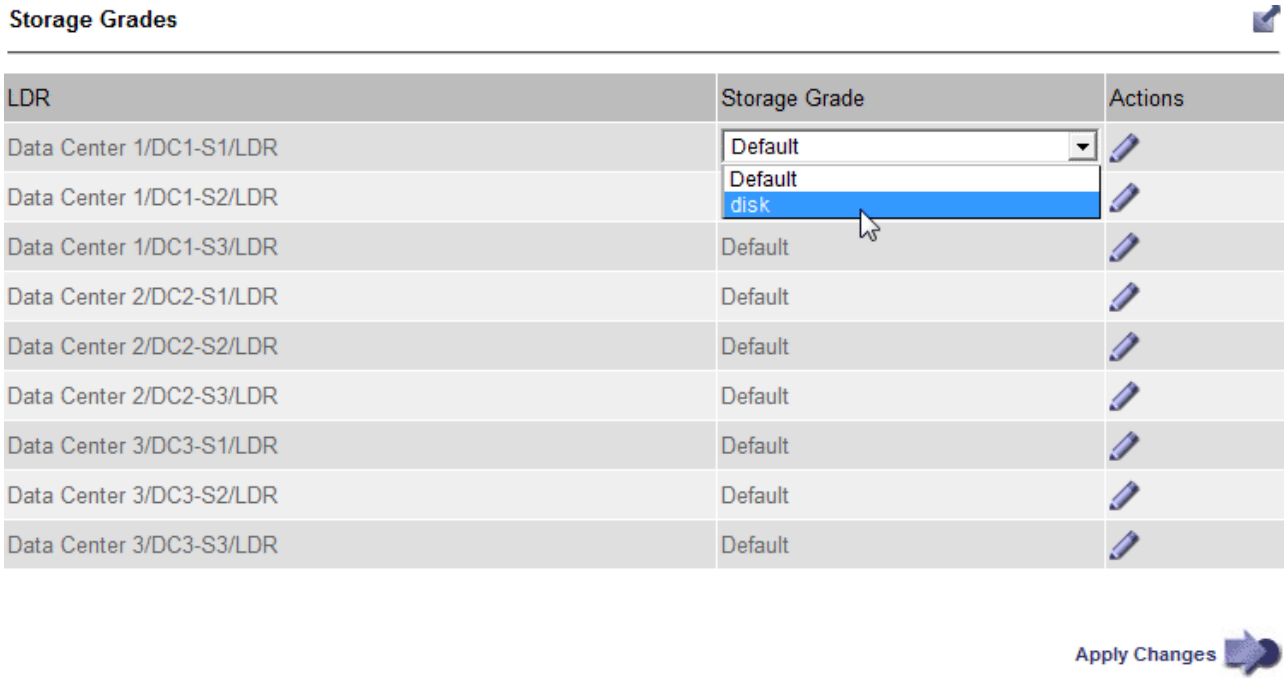
   b. To edit an existing storage grade, click **Edit** and modify the label as required.

      You cannot delete storage grades.

   c. Click **Apply Changes**.

      These storage grades are now available for assignment to Storage Nodes.

3. Assign a storage grade to a Storage Node:
   a. For each Storage Node's LDR service, click **Edit** and select a storage grade from the list.



> Assign a storage grade to a given Storage Node only once. A Storage Node recovered from failure maintains the previously assigned storage grade. Do not change this assignment after the ILM policy is activated. If the assignment is changed, data is stored based on the new storage grade.

b. Click **Apply Changes**.

Configuring storage pools

When defining an ILM rule, you use storage pools to specify where objects are stored . Before creating a storage pool, you must review the storage pool guidelines.

- What a storage pool is
- Guidelines for creating storage pools
- Using multiple storage pools for cross-site replication
- Using a storage pool as a temporary location (deprecated)
- Creating a storage pool
- Viewing storage pool details
- Editing a storage pool
- Removing a storage pool

What a storage pool is

A storage pool is a logical grouping of Storage Nodes or Archive Nodes. You configure storage pools to determine where the StorageGRID system stores object data and the type of storage used.

Storage pools have two attributes:

- **Storage grade**: For Storage Nodes, the relative performance of backing storage.
- **Site**: The data center where objects will be stored.

Storage pools are used in ILM rules to determine where object data is stored. When you configure ILM rules for replication, you select one or more storage pools that include either Storage Nodes or Archive Nodes. When you create Erasure Coding profiles, you select a storage pool that includes Storage Nodes.

Guidelines for creating storage pools

When configuring and using storage pools, follow these guidelines.

Guidelines for all storage pools

- StorageGRID includes a default storage pool, All Storage Nodes, that uses the default site, All Sites, and the default storage grade, All Storage Nodes. The All Storage Nodes storage pool is automatically updated whenever you add new data center sites.

> Using the All Storage Nodes storage pool or the All Sites site is not recommended because these items are automatically updated to include any new sites you add in an expansion, which might not be the behavior you want. Before using the All Storage Nodes storage pool or the default site, carefully review the guidelines for replicated and erasure-coded copies.

- Keep storage pool configurations as simple as possible. Do not create more storage pools than necessary.
- Create storage pools with as many nodes as possible. Each storage pool should contain two or more nodes. A storage pool with insufficient nodes can cause ILM backlogs if a node becomes unavailable.
- Avoid creating or using storage pools that overlap (contain one or more of the same nodes). If storage pools overlap, more than one copy of object data might be saved on the same node.

Guidelines for storage pools used for replicated copies

- Create a different storage pool for each site. Then, specify one or more site-specific storage pools in the placement instructions for each rule. Using a storage pool for each site ensures that replicated object copies are placed exactly where you expect (for example, one copy of every object at each site for site-loss protection).
- If you add a site in an expansion, create a new storage pool for the new site. Then, update ILM rules to control which objects are stored on the new site.
- In general, do not use the default storage pool, All Storage Nodes, or any storage pool that includes the default site, All Sites.

Guidelines for storage pools used for erasure-coded copies

- You cannot use Archive Nodes for erasure-coded data.
- The number of Storage Nodes and sites contained in the storage pool determine which erasure-coding schemes are available.
- If a storage pool includes only two sites, you cannot use that storage pool for erasure coding. No erasure-coding schemes are available for a storage pool that has two sites.
- In general, do not use the default storage pool, All Storage Nodes, or any storage pool that includes the default site, All Sites in any Erasure Coding profile.

> If your grid includes only one site, you are prevented from using the All Storage Nodes storage pool or the All Sites default site in an Erasure Coding profile. This behavior prevents the Erasure Coding profile from becoming invalid if a second site is added.

- If you have high throughput requirements, creating a storage pool that includes multiple sites is not recommended if the network latency between sites is greater than 100 ms. As latency increases, the rate at which StorageGRID can create, place, and retrieve object fragments decreases sharply due to the decrease in TCP network throughput. The decrease in throughput affects the maximum achievable rates of object ingest and retrieval (when Strict or Balanced are selected as the Ingest Behavior) or could lead to ILM queue backlogs (when Dual Commit is selected as the Ingest Behavior).
- If possible, a storage pool should include more than the minimum number of Storage Nodes required for the erasure-coding scheme you select. For example, if you use a 6+3 erasure-coding scheme, you must have at least nine Storage Nodes. However, having at least one additional Storage Node per site is recommended.
- Distribute Storage Nodes across sites as evenly as possible. For example, to support a 6+3 erasure-coding scheme, configure a storage pool that includes at least three Storage Nodes at three sites.

Guidelines for storage pools used for archived copies

- You cannot create a storage pool that includes both Storage Nodes and Archive Nodes. Archived copies require a storage pool that only includes Archive Nodes.
- When using a storage pool that includes Archive Nodes, you should also maintain at least one replicated or erasure-coded copy on a storage pool that includes Storage Nodes.
- If the global S3 Object Lock setting is enabled and you are creating a compliant ILM rule, you cannot use a storage pool that includes Archive Nodes. See the instructions for managing objects with S3 Object Lock.
- If an Archive Node's Target Type is Cloud Tiering - Simple Storage Service (S3), the Archive Node must be in its own storage pool. See the instructions for administering StorageGRID.

Related information

What replication is

What erasure coding is

What erasure-coding schemes are

Using multiple storage pools for cross-site replication

Using a storage pool as a temporary location (deprecated)

Managing objects with S3 Object Lock

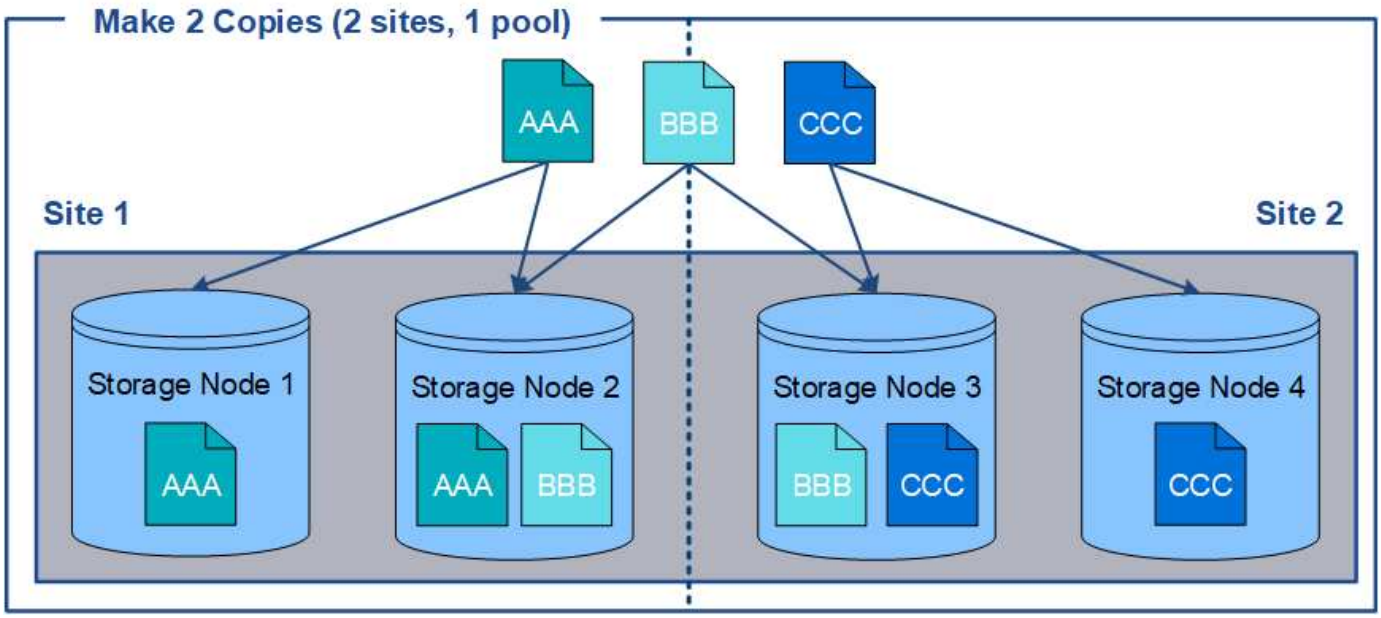
Administer StorageGRID

Using multiple storage pools for cross-site replication

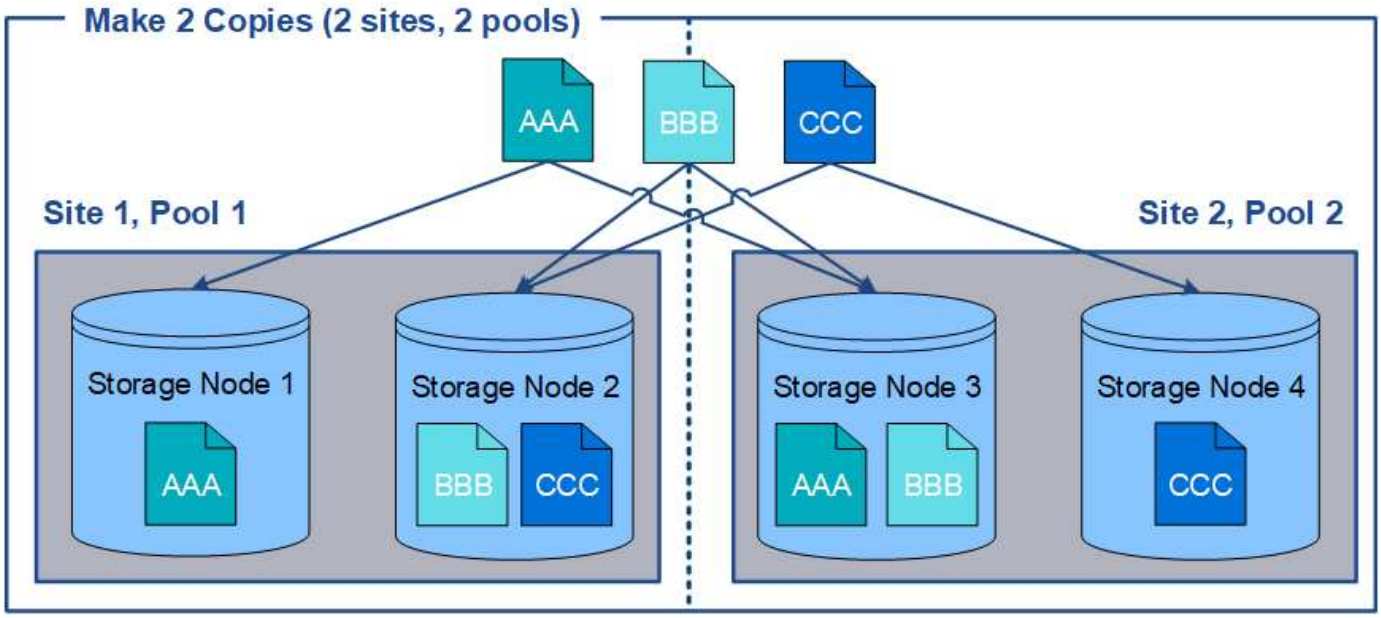
If your StorageGRID deployment includes more than one site, you can enable site-loss protection by creating a storage pool for each site and specifying both storage pools in the rule's placement instructions. For example, if you configure an ILM rule to make two replicated copies and specify storage pools at two sites, one copy of each object will be placed at each site. If you configure a rule to make two copies and specify three storage

pools, the copies are distributed to balance disk usage among the storage pools, while ensuring that the two copies are stored at different sites.

The following example illustrates what can happen if an ILM rule places replicated object copies to a single storage pool containing Storage Nodes from two sites. Because the system uses any available nodes in the storage pool when it places the replicated copies, it might place all copies of some objects within only one of the sites. In this example, the system stored two copies of object AAA on Storage Nodes at Site 1, and two copies of object CCC on Storage Nodes at Site 2. Only object BBB is protected if one of the sites fails or becomes inaccessible.



In contrast, this example illustrates how objects are stored when you use multiple storage pools. In the example, the ILM rule specifies that two replicated copies of each object be created, and that the copies be distributed to two storage pools. Each storage pool contains all Storage Nodes at one site. Because a copy of each object is stored at each site, object data is protected from site failure or inaccessibility.



When using multiple storage pools, keep the following rules in mind:

- If you are creating n copies, you must add n or more storage pools. For example, if a rule is configured to make three copies, you must specify three or more storage pools.
- If the number of copies equals the number of storage pools, one copy of the object is stored in each storage pool.
- If the number of copies is less than the number of storage pools, the system distributes the copies to keep disk usage among the pools balanced and to ensure that two or more copies are not stored in the same storage pool.
- If the storage pools overlap (contain the same Storage Nodes), all copies of the object might be saved at only one site. You must ensure that the selected storage pools do not contain the same Storage Nodes.

Using a storage pool as a temporary location (deprecated)

When you create an ILM rule with an object placement that includes a single storage pool, you are prompted to specify a second storage pool to use as a temporary location.

Temporary locations have been deprecated and will be removed in a future release. You should not select a storage pool as a temporary location for a new ILM rule.

> If you select the Strict ingest behavior (Step 3 of the Create ILM Rule wizard), the temporary location is ignored.

Related information

Data-protection options for ingest

Creating a storage pool

You create storage pools to determine where the StorageGRID system stores object data and the type of storage used. Each storage pool includes one or more sites and one or more storage grades.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have reviewed the guidelines for creating storage pools.

About this task

Storage pools determine where object data is stored. The number of storage pools you need depends on the number of sites in your grid and on the types of copies you want: replicated or erasure-coded.

- For replication and single-site erasure coding, create a storage pool for each site. For example, if you want to store replicated object copies at three sites, create three storage pools.
- For erasure coding at three or more sites, create one storage pool that includes an entry for each site. For example, if you want to erasure code objects across three sites, create one storage pool. Select the plus icon ➕ to add an entry for each site.

> Do not include the default All Sites site in a storage pool that will be used in an Erasure Coding profile. Instead, add a separate entry to the storage pool for each site that will store erasure coded data. See this step for an example.

- If you have more than one storage grade, do not create a storage pool that includes different storage grades at a single site.

  Guidelines for creating storage pools

Steps

1. Select **ILM** > **Storage Pools**.

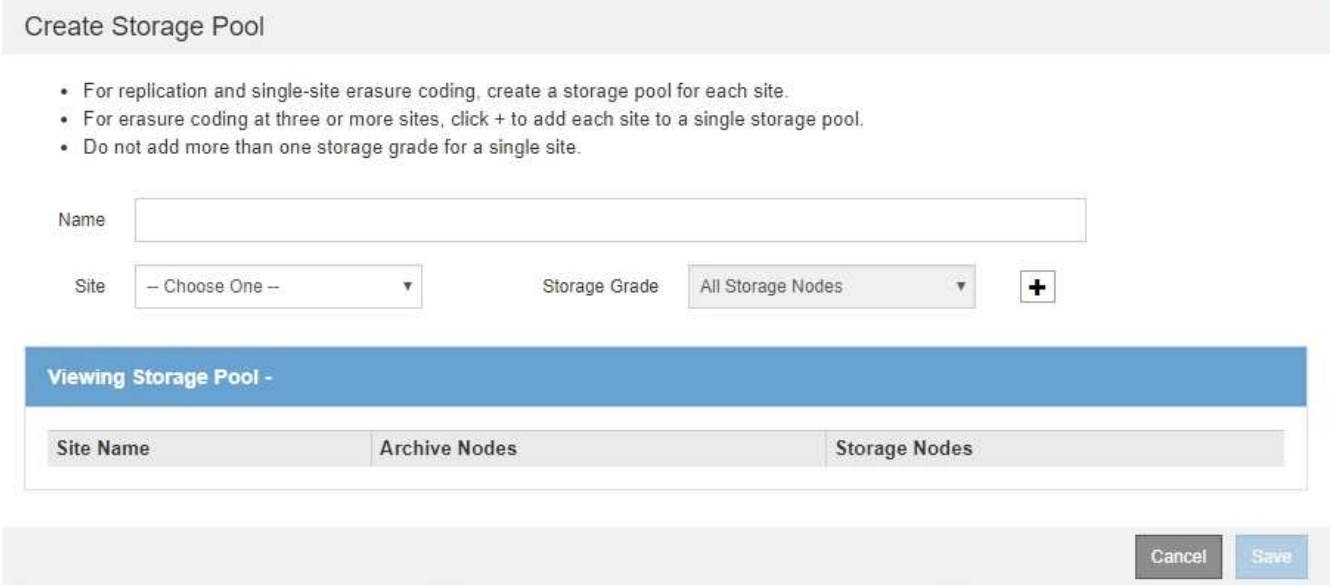
   The Storage Pools page appears and lists all defined storage pools.

   The list includes the system-default storage pool, All Storage Nodes, which uses the system-default site, All Sites, and the default storage grade, All Storage Nodes.

   > Because the All Storage Nodes storage pool is automatically updated whenever you add new data center sites, using this storage pool in ILM rules is not recommended.

2. To create a new storage pool, select **Create**.

   The Create Storage Pool dialog box appears.



3. Enter a unique name for the storage pool.

   Use a name that will be easy to identify when you configure Erasure Coding profiles and ILM rules.

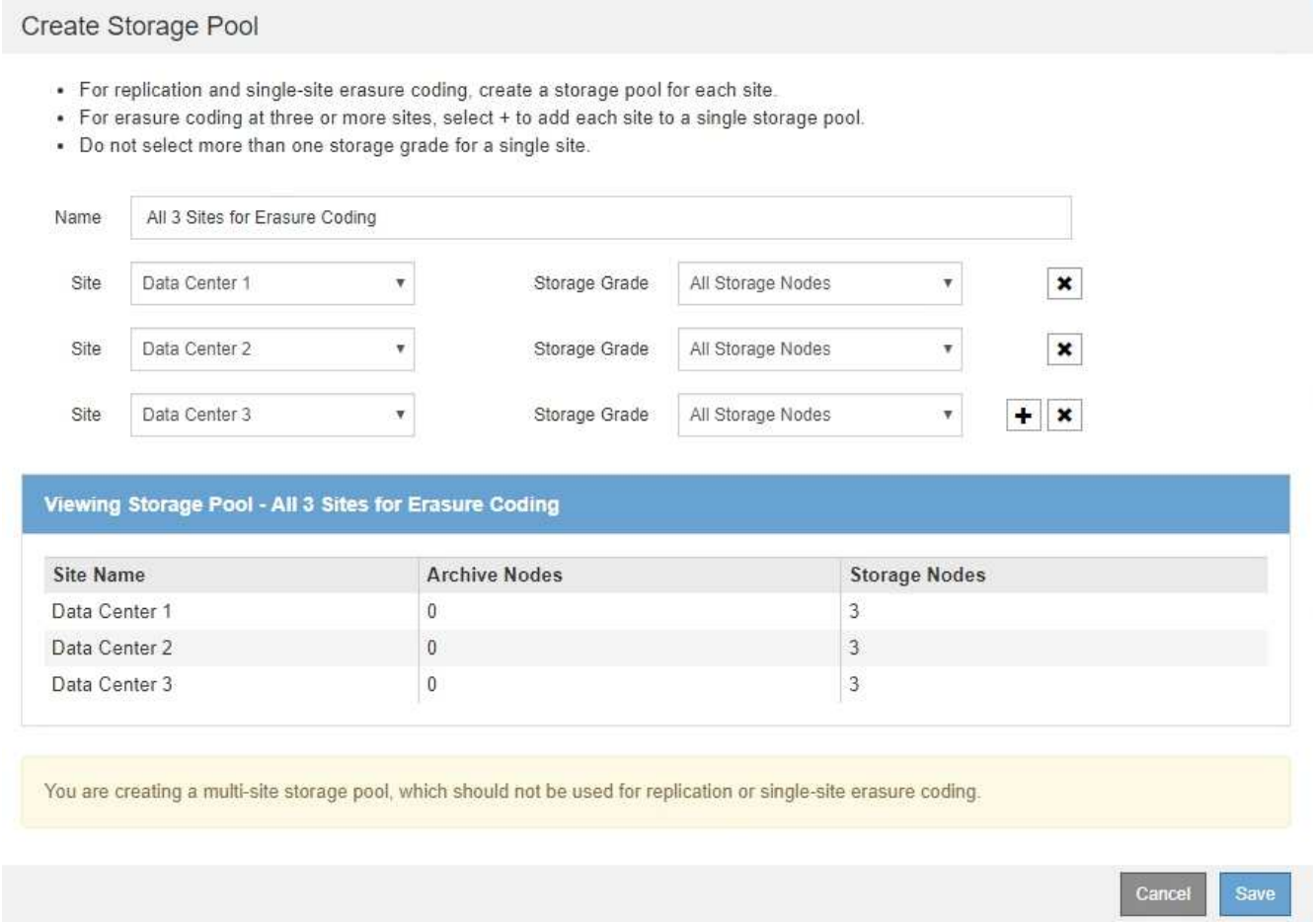
4. From the **Site** drop-down list, select a site for this storage pool.

   When you select a site, the number of Storage Nodes and Archive Nodes in the table are automatically updated.

5. From the **Storage Grade** drop-down list, select the type of storage that will be used if an ILM rule uses this storage pool.

   The default All Storage Nodes storage grade includes all Storage Nodes at the selected site. The default Archive Nodes storage grade includes all Archive Nodes at the selected site. If you created additional storage grades for the Storage Nodes in your grid, they are listed in the drop-down.

6. If you want to use the storage pool in a multi-site Erasure Coding profile, select ➕ to add an entry for each site to the storage pool.



> You are prevented from creating duplicate entries or from creating a storage pool that includes both the **Archive Nodes** storage grade and any storage grade that contains Storage Nodes.
>
> You are warned if you add more than one entry for a site but with different storage grades.

To remove an entry, select ✖.

7. When you are satisfied with your selections, select **Save**.

   The new storage pool is added to the list.

Related information

Guidelines for creating storage pools

Viewing storage pool details

You can view the details of a storage pool to determine where the storage pool is used and to see which nodes and storage grades are included.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.

Steps

1. Select **ILM** > **Storage Pools**.

   The Storage Pools page appears. This page lists all defined storage pools.

   Storage Pools

   Storage Pools

   A storage pool is a logical group of Storage Nodes or Archive Nodes and is used in ILM rules to determine where object data is stored.

   + Create | Edit | ✖ Remove | View Details

Name	Used Space	Free Space	Total Capacity	ILM Usage
All Storage Nodes	1.88 MB	2.80 TB	2.80 TB	Used in 1 ILM rule
DC1	621.77 KB	932.42 GB	932.42 GB	Used in 2 ILM rules
DC2	675.82 KB	932.42 GB	932.42 GB	Used in 2 ILM rules
DC3	578.95 KB	932.42 GB	932.42 GB	Used in 1 ILM rule
All 3 Sites	1.88 MB	2.80 TB	2.80 TB	Used in 1 ILM rule and 1 EC profile
Archive	—	—	—	—

   Displaying 6 storage pools.

   Cloud Storage Pools

   You can add Cloud Storage Pools to ILM rules to store objects outside of the StorageGRID system. A Cloud Storage Pool defines how to access the external bucket or container where objects will be stored.

   + Create | Edit | ✖ Remove | Clear Error

   No Cloud Storage Pools found.

   The table includes the following information for each storage pool that includes Storage Nodes:

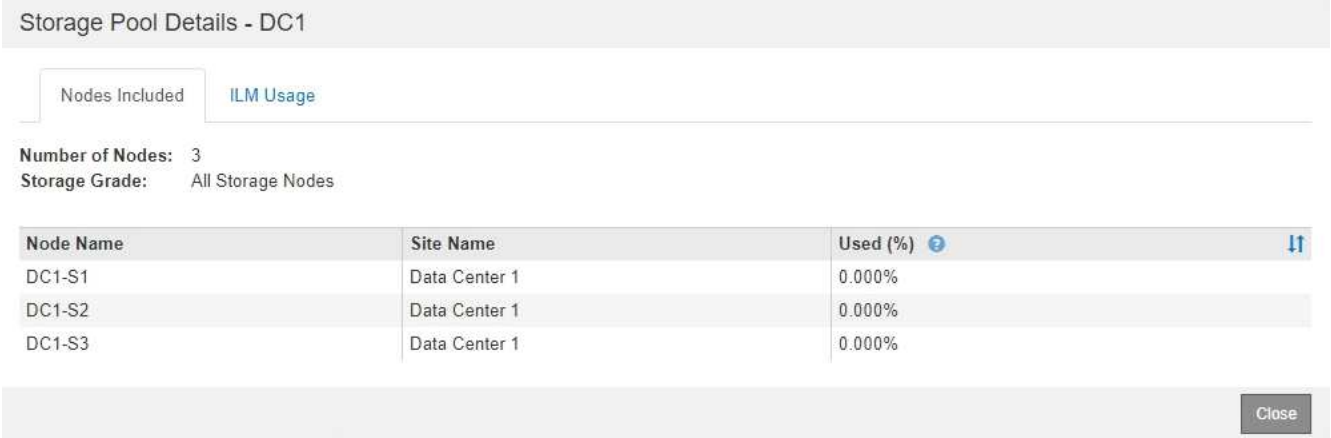
   - **Name**: The unique display name of the storage pool.
   - **Used Space**: The amount of space that is currently being used to store objects in the storage pool.
   - **Free Space**: The amount of space that remains available to store objects in the storage pool.
   - **Total Capacity**: The size of the storage pool, which equals the total amount of usable space for object data for all nodes in the storage pool .
   - **ILM Usage**: How the storage pool is currently being used. A storage pool might be unused or it might be used in one or more ILM rules, Erasure Coding profiles, or both.

     You cannot remove a storage pool if it is being used.

2. To view details about a specific storage pool, select its radio button and select **View Details**.

   The Storage Pool Details modal appears.

3. View the **Nodes Included** tab to learn about the Storage Nodes or Archive Nodes included in the storage pool.



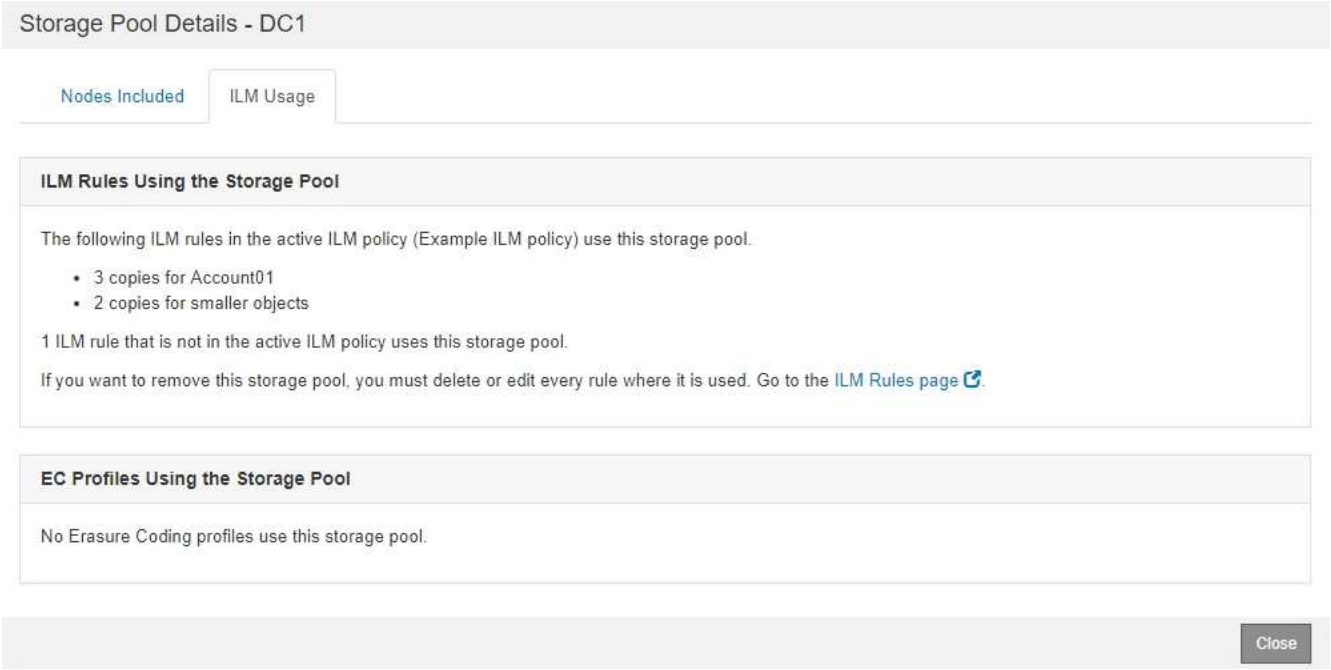
The table includes the following information for each node:

- Node Name
- Site Name
- Used (%): For Storage Nodes, the percentage of the total usable space for object data that has been used. This value does not include object metadata.

> The same Used (%) value is also shown in the Storage Used - Object Data chart for each Storage Node (select **Nodes** > ***Storage Node*** > **Storage**).

4. Select the **ILM Usage** tab to determine if the storage pool is currently being used in any ILM rules or Erasure Coding profiles.

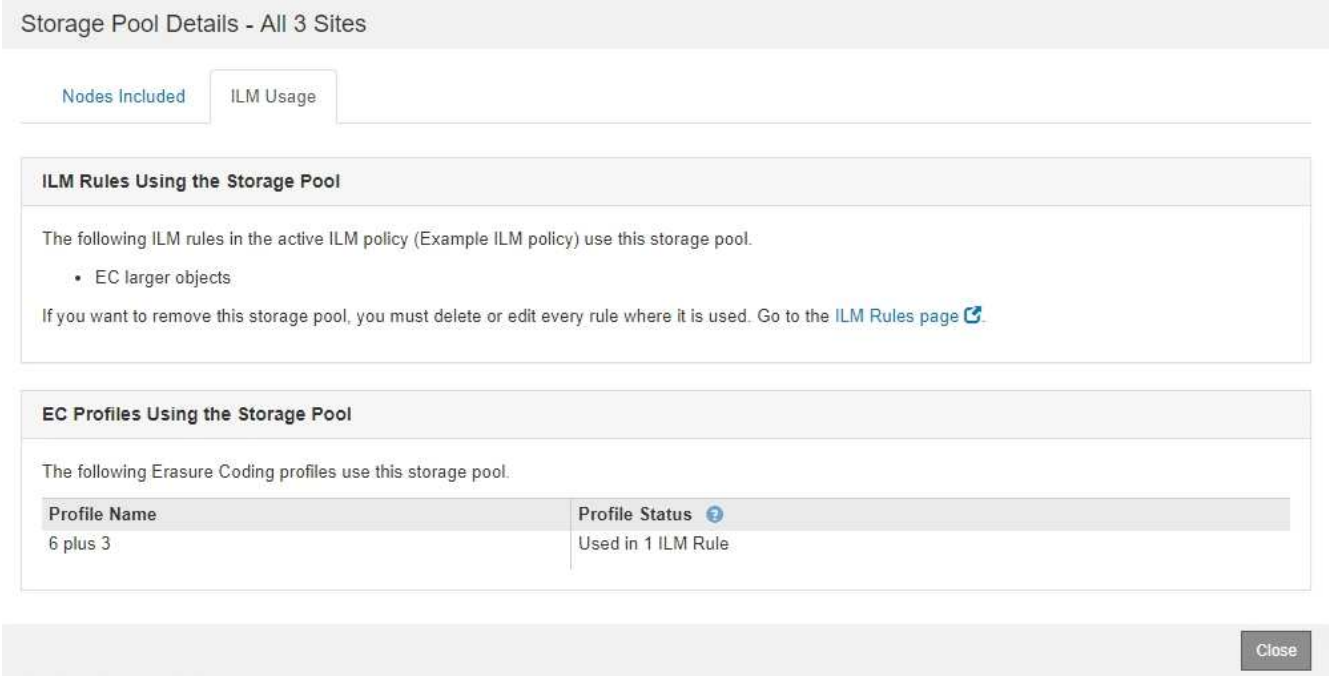
   In this example, the DC1 storage pool is used in three ILM rules: two rules that are in the active ILM policy and one rule that is not in the active policy.



> You cannot remove a storage pool if it is used in an ILM rule.

In this example, the All 3 Sites storage pool is used in an Erasure Coding profile. In turn, that Erasure

Coding profile is used by one ILM rule in the active ILM policy.



You cannot remove a storage pool if it is used in an Erasure Coding profile.

5. Optionally, go to the **ILM Rules page** to learn about and manage any rules that use the storage pool.

   See the instructions for working with ILM rules.

6. When you are done viewing storage pool details, select **Close**.

Related information

Working with ILM rules and ILM policies

Editing a storage pool

You can edit a storage pool to change its name or to update sites and storage grades.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have reviewed the guidelines for creating storage pools.
- If you plan to edit a storage pool that is used by a rule in the active ILM policy, you must have considered how your changes will affect object data placement.

About this task

If you are adding a new storage grade to a storage pool that is used in the active ILM policy, be aware that the Storage Nodes in the new storage grade will not be used automatically. To force StorageGRID to use a new storage grade, you must activate a new ILM policy after saving the edited storage pool.

Steps

1. Select **ILM** > **Storage Pools**.

The Storage Pools page appears.

2. Select the radio button for the storage pool you want to edit.

   You cannot edit the All Storage Nodes storage pool.

3. Select **Edit**.
4. As required, change the storage pool name.
5. As required, select other sites and storage grades.

   > You are prevented from changing the site or storage grade if the storage pool is used in an Erasure Coding profile and the change would cause the erasure-coding scheme to become invalid. For example, if a storage pool used in a Erasure Coding profile currently includes a storage grade with only one site, you are prevented from using a storage grade with two sites since the change would make the erasure-coding scheme invalid.

6. Select **Save**.

After you finish

If you added a new storage grade to a storage pool used in the active ILM policy, activate a new ILM policy to force StorageGRID to use the new storage grade. For example, clone your existing ILM policy and then activate the clone.

Removing a storage pool

You can remove a storage pool that is not being used.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.

Steps

1. Select **ILM** > **Storage Pools**.

   The Storage Pools page appears.

2. Look at the ILM Usage column in the table to determine whether you can remove the storage pool.

   You cannot remove a storage pool if it is being used in an ILM rule or in an Erasure Coding profile. As required, select **View Details** > **ILM Usage** to determine where a storage pool is used.

3. If the storage pool you want to remove is not being used, select the radio button.
4. Select **Remove**.
5. Select **OK**.

Using Cloud Storage Pools

You can use Cloud Storage Pools to move StorageGRID objects to an external storage location, such as S3 Glacier or Microsoft Azure Blob storage. Moving objects outside of the grid lets you take advantage of a low-cost storage tier for long-term archive.

- What a Cloud Storage Pool is
- Lifecycle of a Cloud Storage Pool object
- When to use Cloud Storage Pools
- Considerations for Cloud Storage Pools
- Comparing Cloud Storage Pools and CloudMirror replication
- Creating a Cloud Storage Pool
- Editing a Cloud Storage Pool
- Removing a Cloud Storage Pool
- Troubleshooting Cloud Storage Pools

What a Cloud Storage Pool is

A Cloud Storage Pool lets you use ILM to move object data outside of your StorageGRID system. For example, you might want to move infrequently accessed objects to lower-cost cloud storage, such as Amazon S3 Glacier, S3 Glacier Deep Archive, or the Archive access tier in Microsoft Azure Blob storage. Or, you might want to maintain a cloud backup of StorageGRID objects to enhance disaster recovery.

From an ILM perspective, a Cloud Storage Pool is similar to a storage pool. To store objects in either location, you select the pool when creating the placement instructions for an ILM rule. However, while storage pools consist of Storage Nodes or Archive Nodes within the StorageGRID system, a Cloud Storage Pool consists of an external bucket (S3) or container (Azure Blob storage).

The following table compares storage pools to Cloud Storage Pools and shows the high-level similarities and differences.

	Storage pool	**Cloud Storage Pool**
How is it created?	Using the **ILM** > **Storage Pools** option in Grid Manager. You must set up storage grades before you can create the storage pool.	Using the **ILM** > **Storage Pools** option in Grid Manager. You must set up the external bucket or container before you can create the Cloud Storage Pool.
How many pools can you create?	Unlimited.	Up to 10.

	Storage pool	Cloud Storage Pool
Where are objects stored?	On one or more Storage Nodes or Archive Nodes within StorageGRID.	In an Amazon S3 bucket or Azure Blob storage container that is external to the StorageGRID system. If the Cloud Storage Pool is an Amazon S3 bucket: • You can optionally configure a bucket lifecycle to transition objects to low-cost, long-term storage, such as Amazon S3 Glacier or S3 Glacier Deep Archive. The external storage system must support the Glacier storage class and the S3 POST Object restore API. • You can create Cloud Storage Pools for use with AWS Commercial Cloud Services (C2S), which supports the AWS Secret Region. If the Cloud Storage Pool is an Azure Blob storage container, StorageGRID transitions the object to the Archive tier. **Note:** In general, do not configure Azure Blob Storage lifecycle management for the container used for a Cloud Storage Pool. POST Object restore operations on objects in the Cloud Storage Pool can be affected by the configured lifecycle.
What controls object placement?	An ILM rule in the active ILM policy.	An ILM rule in the active ILM policy.
What data protection method is used?	Replication or erasure coding.	Replication.
How many copies of each object are allowed?	Multiple.	One copy in the Cloud Storage Pool and, optionally, one or more copies in StorageGRID. **Note:** You cannot store an object in more than one Cloud Storage Pool at any given time.
What are the advantages?	Objects are quickly accessible at any time.	Low-cost storage.

Lifecycle of a Cloud Storage Pool object

Before implementing Cloud Storage Pools, review the lifecycle of objects that are stored in each type of Cloud Storage Pool.

Related information

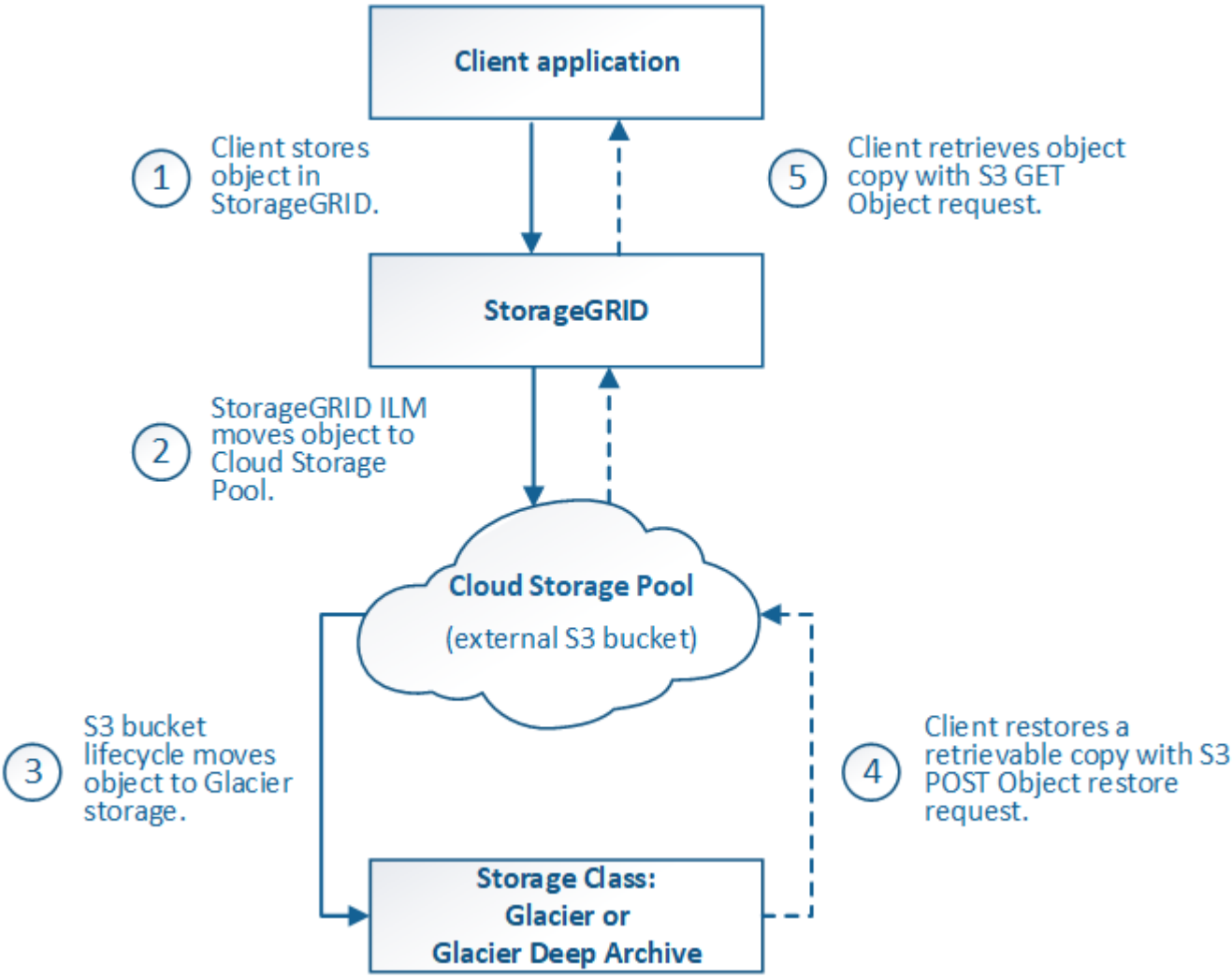
S3: Lifecycle of a Cloud Storage Pool object

Azure: Lifecycle of a Cloud Storage Pool object]

S3: Lifecycle of a Cloud Storage Pool object

The figure shows the lifecycle stages of an object that is stored in an S3 Cloud Storage Pool.

> In the figure and explanations, “Glacier” refers to both the Glacier storage class and the Glacier Deep Archive storage class, with one exception: the Glacier Deep Archive storage class does not support the Expedited restore tier. Only Bulk or Standard retrieval is supported.



1. **Object stored in StorageGRID**

   To start the lifecycle, a client application stores an object in StorageGRID.

2. **Object moved to S3 Cloud Storage Pool**
   - When the object is matched by an ILM rule that uses an S3 Cloud Storage Pool as its placement location, StorageGRID moves the object to the external S3 bucket specified by the Cloud Storage Pool.
   - When the object has been moved to the S3 Cloud Storage Pool, the client application can retrieve it using an S3 GET Object request from StorageGRID, unless the object has been transitioned to Glacier storage.
3. **Object transitioned to Glacier (non-retrievable state)**
   - Optionally, the object can be transitioned to Glacier storage. For example, the external S3 bucket might use lifecycle configuration to transition an object to Glacier storage immediately or after some number of days.

     > If you want to transition objects, you must create a lifecycle configuration for the external S3 bucket, and you must use a storage solution that implements the Glacier storage class and supports the S3 POST Object restore API.

     > Do not use Cloud Storage Pools for objects that have been ingested by Swift clients. Swift does not support POST Object restore requests, so StorageGRID will not be able to retrieve any Swift objects that have been transitioned to S3 Glacier storage. Issuing a Swift GET object request to retrieve these objects will fail (403 Forbidden).

   - During the transition, the client application can use an S3 HEAD Object request to monitor the object's status.
4. **Object restored from Glacier storage**

   If an object has been transitioned to Glacier storage, the client application can issue an S3 POST Object restore request to restore a retrievable copy to the S3 Cloud Storage Pool. The request specifies how many days the copy should be available in the Cloud Storage Pool and the data-access tier to use for the restore operation (Expedited, Standard, or Bulk). When the expiration date of the retrievable copy is reached, the copy is automatically returned to a non-retrievable state.

   > If one or more copies of the object also exist on Storage Nodes within StorageGRID, there is no need to restore the object from Glacier by issuing a POST Object restore request. Instead, the local copy can be retrieved directly, using a GET Object request.

5. **Object retrieved**

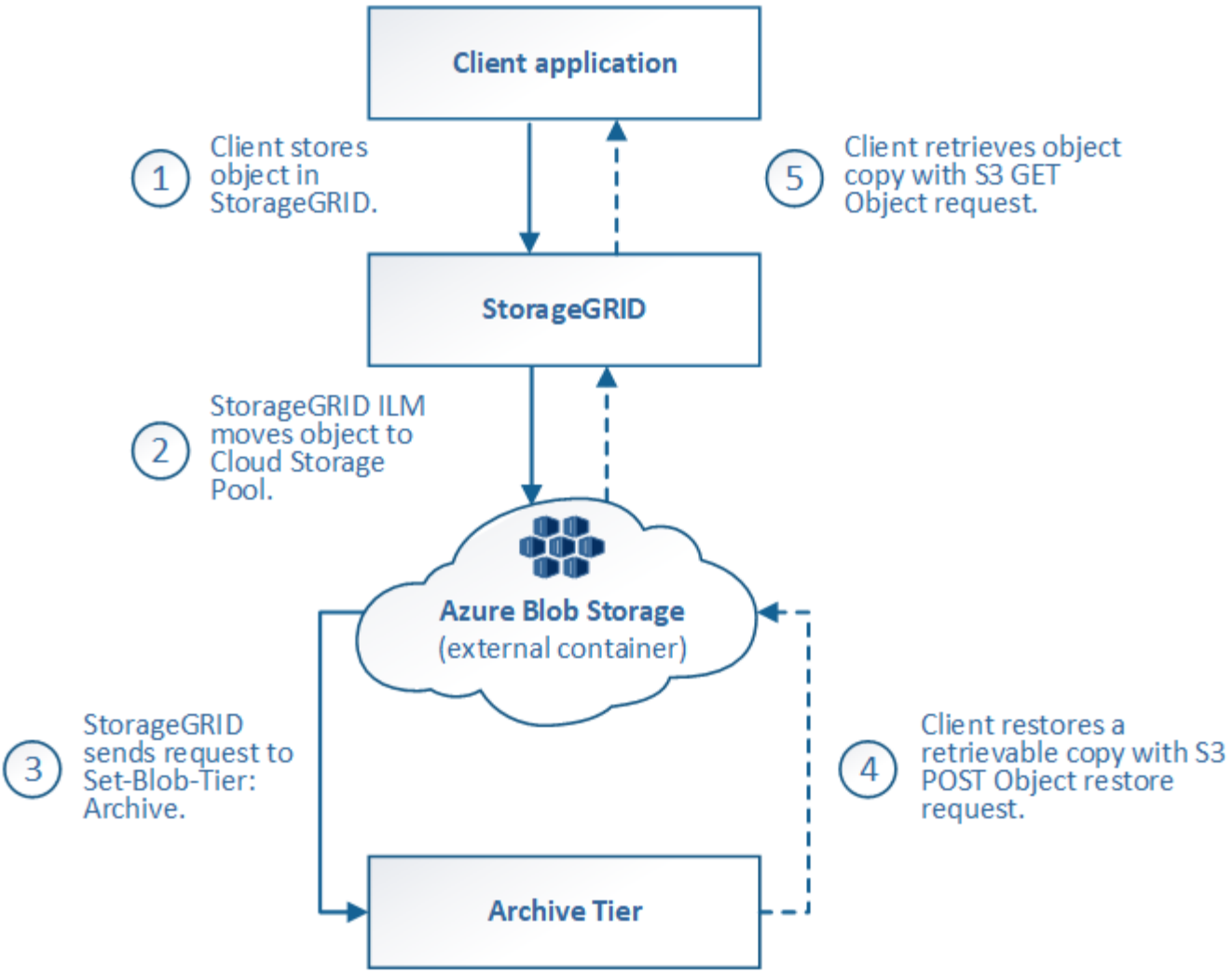
   Once an object has been restored, the client application can issue a GET Object request to retrieve the restored object.

Related information

Use S3

Azure: Lifecycle of a Cloud Storage Pool object

The figure shows the lifecycle stages of an object that is stored in an Azure Cloud Storage Pool.



1. **Object stored in StorageGRID**

   To start the lifecycle, a client application stores an object in StorageGRID.

2. **Object moved to Azure Cloud Storage Pool**

   When the object is matched by an ILM rule that uses an Azure Cloud Storage Pool as its placement location, StorageGRID moves the object to the external Azure Blob storage container specified by the Cloud Storage Pool

   > Do not use Cloud Storage Pools for objects that have been ingested by Swift clients. Swift does not support POST Object restore requests, so StorageGRID will not be able to retrieve any Swift objects that have been transitioned to the Azure Blob storage Archive tier. Issuing a Swift GET object request to retrieve these objects will fail (403 Forbidden).

3. **Object transitioned to Archive tier (non-retrievable state)**

   Immediately after moving the object to the Azure Cloud Storage Pool, StorageGRID automatically transitions the object to the Azure Blob storage Archive tier.

4. **Object restored from Archive tier**

   If an object has been transitioned to the Archive tier, the client application can issue an S3 POST Object

restore request to restore a retrievable copy to the Azure Cloud Storage Pool.

When StorageGRID receives the POST Object Restore, it temporarily transitions the object to the Azure Blob storage Cool tier. As soon as the expiration date in the POST Object restore request is reached, StorageGRID transitions the object back to the Archive tier.

> If one or more copies of the object also exist on Storage Nodes within StorageGRID, there is no need to restore the object from the Archive access tier by issuing a POST Object restore request. Instead, the local copy can be retrieved directly, using a GET Object request.

5. **Object retrieved**

   Once an object has been restored to the Azure Cloud Storage Pool, the client application can issue a GET Object request to retrieve the restored object.

When to use Cloud Storage Pools

Cloud Storage Pools can provide significant benefits in several use cases.

Backing up StorageGRID data in an external location

You can use a Cloud Storage Pool to back up StorageGRID objects to an external location.

If the copies in StorageGRID are inaccessible, the object data in the Cloud Storage Pool can be used to serve client requests. However, you might need to issue S3 POST Object restore request to access the backup object copy in the Cloud Storage Pool.

The object data in a Cloud Storage Pool can also be used to recover data lost from StorageGRID because of a storage volume or Storage Node failure. If the only remaining copy of an object is in a Cloud Storage Pool, StorageGRID temporarily restores the object and creates a new copy on the recovered Storage Node.

To implement a backup solution:

1. Create a single Cloud Storage Pool.
2. Configure an ILM rule that simultaneously stores object copies on Storage Nodes (as replicated or erasure-coded copies) and a single object copy in the Cloud Storage Pool.
3. Add the rule to your ILM policy. Then, simulate and activate the policy.

Tiering data from StorageGRID to external location

You can use a Cloud Storage Pool to store objects outside of the StorageGRID system. For example, suppose you have a large number of objects that you need to retain, but you expect to access those objects rarely, if ever. You can use a Cloud Storage Pool to tier the objects to lower-cost storage and to free up space in StorageGRID.

To implement a tiering solution:

1. Create a single Cloud Storage Pool.
2. Configure an ILM rule that moves rarely used objects from Storage Nodes to the Cloud Storage Pool.
3. Add the rule to your ILM policy. Then, simulate and activate the policy.

Maintain multiple cloud endpoints

You can configure multiple Cloud Storage Pools if you want to tier or back up object data to more than one cloud. The filters in your ILM rules let you specify which objects are stored in each Cloud Storage Pool. For example, you might want to store objects from some tenants or buckets in Amazon S3 Glacier and objects from other tenant or buckets in Azure Blob storage. Or, you might want to move data between Amazon S3 Glacier and Azure Blob storage. When using multiple Cloud Storage Pools, keep in mind that an object can be stored in only one Cloud Storage Pool at a time.

To implement multiple cloud endpoints:

1. Create up to 10 Cloud Storage Pools.
2. Configure ILM rules to store the appropriate object data at the appropriate time in each Cloud Storage Pool. For example, store objects from bucket A in Cloud Storage Pool A, and store objects from bucket B in Cloud Storage Pool B. Or, store objects in Cloud Storage Pool A for some amount of time and then move them to Cloud Storage Pool B.
3. Add the rules to your ILM policy. Then, simulate and activate the policy.

Considerations for Cloud Storage Pools

If you plan to use a Cloud Storage Pool to move objects out of the StorageGRID system, you must review the considerations for configuring and using Cloud Storage Pools.

General considerations

- In general, cloud archival storage, such as Amazon S3 Glacier or Azure Blob storage, is an inexpensive place to store object data. However, the costs to retrieve data from cloud archival storage are relatively high. To achieve the lowest overall cost, you must consider when and how often you will access the objects in the Cloud Storage Pool. Using a Cloud Storage Pool is recommended only for content that you expect to access infrequently.
- Do not use Cloud Storage Pools for objects that have been ingested by Swift clients. Swift does not support POST Object restore requests, so StorageGRID will not be able to retrieve any Swift objects that have been transitioned to S3 Glacier storage or the Azure Blob storage Archive tier. Issuing a Swift GET object request to retrieve these objects will fail (403 Forbidden).
- Using Cloud Storage Pools with FabricPool is not supported because of the added latency to retrieve an object from the Cloud Storage Pool target.

Information required to create a Cloud Storage Pool

Before you can create a Cloud Storage Pool, you must create the external S3 bucket or the external Azure Blob storage container that you will use for the Cloud Storage Pool. Then, when you create the Cloud Storage Pool in StorageGRID, you must specify the following information:

- The provider type: Amazon S3 or Azure Blob storage.
- If you select Amazon S3, whether the Cloud Storage Pool is for use with the AWS Secret Region (**CAP (C2S Access Portal)**).
- The exact name of the bucket or container.
- The service endpoint needed to access the bucket or container.
- The authentication needed to access the bucket or container:
  - **S3**: Optionally, an access key ID and secret access key.

- **C2S**: The complete URL for obtaining temporary credentials from the CAP server; a server CA certificate, a client certificate, a private key for the client certificate, and, if the private key is encrypted, the passphrase for decrypting it.
- **Azure Blob storage**: An account name and account key. These credentials must have full permission for the container.

- Optionally, a custom CA certificate to verify TLS connections to the bucket or container.

Considerations for the ports used for Cloud Storage Pools

To ensure that the ILM rules can move objects to and from the specified Cloud Storage Pool, you must configure the network or networks that contain your system's Storage Nodes. You must ensure that the following ports can communicate with the Cloud Storage Pool.

By default, Cloud Storage Pools use the following ports:

- **80**: For endpoint URIs that begin with http
- **443**: For endpoint URIs that begin with https

You can specify a different port when you create or edit a Cloud Storage Pool.

If you use a non-transparent proxy server, you must also configure a Storage proxy to allow messages to be sent to external endpoints, such as an endpoint on the internet.

Considerations for costs

Access to storage in the cloud using a Cloud Storage Pool requires network connectivity to the cloud. You must consider the cost of the network infrastructure you will use to access the cloud and provision it appropriately, based on the amount of data you expect to move between StorageGRID and the cloud using the Cloud Storage Pool.

When StorageGRID connects to the external Cloud Storage Pool endpoint, it issues various requests to monitor connectivity and to ensure it can perform the required operations. While some additional costs will be associated with these requests, the cost of monitoring a Cloud Storage Pool should only be a small fraction of the overall cost of storing objects in S3 or Azure.

More significant costs might be incurred if you need to move objects from an external Cloud Storage Pool endpoint back to StorageGRID. Objects might be moved back to StorageGRID in either of these cases:

- The only copy of the object is in a Cloud Storage Pool and you decide to store the object in StorageGRID instead. In this case, you simply reconfigure your ILM rules and policy. When ILM evaluation occurs, StorageGRID issues multiple requests to retrieve the object from the Cloud Storage Pool. StorageGRID then creates the specified number of replicated or erasure-coded copies locally. After the object is moved back to StorageGRID, the copy in the Cloud Storage Pool is deleted.
- Objects are lost because of Storage Node failure. If the only remaining copy of an object is in a Cloud Storage Pool, StorageGRID temporarily restores the object and creates a new copy on the recovered Storage Node.

> When objects are moved back to StorageGRID from a Cloud Storage Pool, StorageGRID issues multiple requests to the Cloud Storage Pool endpoint for each object. Before moving large numbers of objects, contact technical support for help in estimating the time frame and associated costs.

S3: Permissions required for the Cloud Storage Pool bucket

The bucket policy for the external S3 bucket used for a Cloud Storage Pool must grant StorageGRID permission to move an object to the bucket, get an object's status, restore an object from Glacier storage when required, and more. Ideally, StorageGRID should have full-control access to the bucket (`s3:*`); however, if this is not possible, the bucket policy must grant the following S3 permissions to StorageGRID:

- `s3:AbortMultipartUpload`
- `s3:DeleteObject`
- `s3:GetObject`
- `s3:ListBucket`
- `s3:ListBucketMultipartUploads`
- `s3:ListMultipartUploadParts`
- `s3:PutObject`
- `s3:RestoreObject`

S3: Considerations for the external bucket's lifecycle

The movement of objects between StorageGRID and the external S3 bucket specified in the Cloud Storage Pool is controlled by ILM rules and the active ILM policy in StorageGRID. In contrast, the transition of objects from the external S3 bucket specified in the Cloud Storage Pool to Amazon S3 Glacier or S3 Glacier Deep Archive (or to a storage solution that implements the Glacier storage class) is controlled by that bucket's lifecycle configuration.

If you want to transition objects from the Cloud Storage Pool, you must create the appropriate lifecycle configuration on the external S3 bucket, and you must use a storage solution that implements the Glacier storage class and supports the S3 POST Object restore API.

For example, suppose you want all objects that are moved from StorageGRID to the Cloud Storage Pool to be transitioned to Amazon S3 Glacier storage immediately. You would create a lifecycle configuration on the external S3 bucket that specifies a single action (**Transition**) as follows:

```
<LifecycleConfiguration>
  <Rule>
    <ID>Transition Rule</ID>
    <Filter>
       <Prefix></Prefix>
    </Filter>
    <Status>Enabled</Status>
    <Transition>
      <Days>0</Days>
      <StorageClass>GLACIER</StorageClass>
    </Transition>
  </Rule>
</LifecycleConfiguration>
```

This rule would transition all bucket objects to Amazon S3 Glacier on the day they were created (that is, on the day they were moved from StorageGRID to the Cloud Storage Pool).

> When configuring the external bucket's lifecycle, never use **Expiration** actions to define when objects expire. Expiration actions cause the external storage system to delete expired objects. If you later attempt to access an expired object from StorageGRID, the deleted object will not be found.

If you want to transition objects in the Cloud Storage Pool to S3 Glacier Deep Archive (instead of to Amazon S3 Glacier), specify `<StorageClass>DEEP_ARCHIVE</StorageClass>` in the bucket lifecycle. However, be aware that you cannot use the `Expedited` tier to restore objects from S3 Glacier Deep Archive.

Azure: Considerations for Access tier

When you configure an Azure storage account, you can set the default Access tier to Hot or Cool. When creating a storage account for use with a Cloud Storage Pool, you should use the Hot tier as the default tier. Even though StorageGRID immediately sets the tier to Archive when it moves objects to the Cloud Storage Pool, using a default setting of Hot ensures that you will not be charged an early deletion fee for objects removed from the Cool tier before the 30-day minimum.

Azure: Lifecycle management not supported

Do not use Azure Blob Storage lifecycle management for the container used with a Cloud Storage Pool. The lifecycle operations might interfere with Cloud Storage Pool operations.

Related information

Creating a Cloud Storage Pool

S3: Specifying authentication details for a Cloud Storage Pool

C2S S3: Specifying authentication details for a Cloud Storage Pool

Azure: Specifying authentication details for a Cloud Storage Pool

Administer StorageGRID

Comparing Cloud Storage Pools and CloudMirror replication

As you begin using Cloud Storage Pools, it might be helpful to understand the similarities and differences between Cloud Storage Pools and the StorageGRID CloudMirror replication service.

	Cloud Storage Pool	CloudMirror replication service
What is the primary purpose?	A Cloud Storage Pool acts as an archive target. The object copy in the Cloud Storage Pool can be the only copy of the object, or it can be an additional copy. That is, instead of keeping two copies on-premise, you can keep only one copy within StorageGRID and send a copy to the Cloud Storage Pool.	The CloudMirror replication service enables a tenant to automatically replicate objects from a bucket in StorageGRID (source) to an external S3 bucket (destination). CloudMirror replication creates an independent copy of an object in an independent S3 infrastructure.

	Cloud Storage Pool	CloudMirror replication service
How is it set up?	Cloud Storage Pools are defined in the same way as storage pools, using the Grid Manager or the Grid Management API. A Cloud Storage Pool can be selected as the placement location in an ILM rule. While a storage pool consists of a group of Storage Nodes, a Cloud Storage Pool is defined using a remote S3 or Azure endpoint (IP address, credentials, and so on).	A tenant user configures CloudMirror replication by defining a CloudMirror endpoint (IP address, credentials, and so on) using the Tenant Manager or the S3 API. After the CloudMirror endpoint is set up, any bucket owned by that tenant account can be configured to point to the CloudMirror endpoint.
Who is responsible for setting it up?	Typically, a grid administrator	Typically, a tenant user
What is the destination?	• Any compatible S3 infrastructure (including Amazon S3) • Azure Blob Archive tier	• Any compatible S3 infrastructure (including Amazon S3)
What causes objects to be moved to the destination?	One or more ILM rules in the active ILM policy. The ILM rules define which objects StorageGRID moves to the Cloud Storage Pool and when the objects are moved.	The act of ingesting a new object into a source bucket that has been configured with a CloudMirror endpoint.Objects that existed in the source bucket before the bucket was configured with the CloudMirror endpoint are not replicated, unless they are modified.
How are objects retrieved?	Applications must make requests to StorageGRID to retrieve objects that have been moved to a Cloud Storage Pool. If the only copy of an object has been transitioned to archival storage, StorageGRID manages the process of restoring the object so it can be retrieved.	Because the mirrored copy in the destination bucket is an independent copy, applications can retrieve the object by making requests either to StorageGRID or to the S3 destination. For example, suppose you use CloudMirror replication to mirror objects to a partner organization. The partner can use its own applications to read or update objects directly from the S3 destination. Using StorageGRID is not required.
Can you read from the destination directly?	No. Objects moved to a Cloud Storage Pool are managed by StorageGRID. Read requests must be directed to StorageGRID (and StorageGRID will be responsible for retrieval from Cloud Storage Pool).	Yes, because the mirrored copy is an independent copy.
What happens if an object is deleted from the source?	The object is also deleted in the Cloud Storage Pool.	The delete action is not replicated. A deleted object no longer exists in the StorageGRID bucket, but it continues to exist in the destination bucket. Similarly, objects in the destination bucket can be deleted without affecting the source.

	Cloud Storage Pool	CloudMirror replication service
How do you access objects after a disaster (StorageGRID system not operational)?	Failed StorageGRID nodes must be recovered. During this process, copies of replicated objects might be restored using the copies in the Cloud Storage Pool.	The object copies in the CloudMirror destination are independent of StorageGRID, so they can be accessed directly before the StorageGRID nodes are recovered.

Related information

Administer StorageGRID

Creating a Cloud Storage Pool

When you create a Cloud Storage Pool, you specify the name and location of the external bucket or container that StorageGRID will use to store objects, the cloud provider type (Amazon S3 or Azure Blob Storage), and the information StorageGRID needs to access the external bucket or container.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have reviewed the guidelines for configuring Cloud Storage Pools.
- The external bucket or container referenced by the Cloud Storage Pool must exist.
- You must have all of the authentication information needed to access the bucket or container.

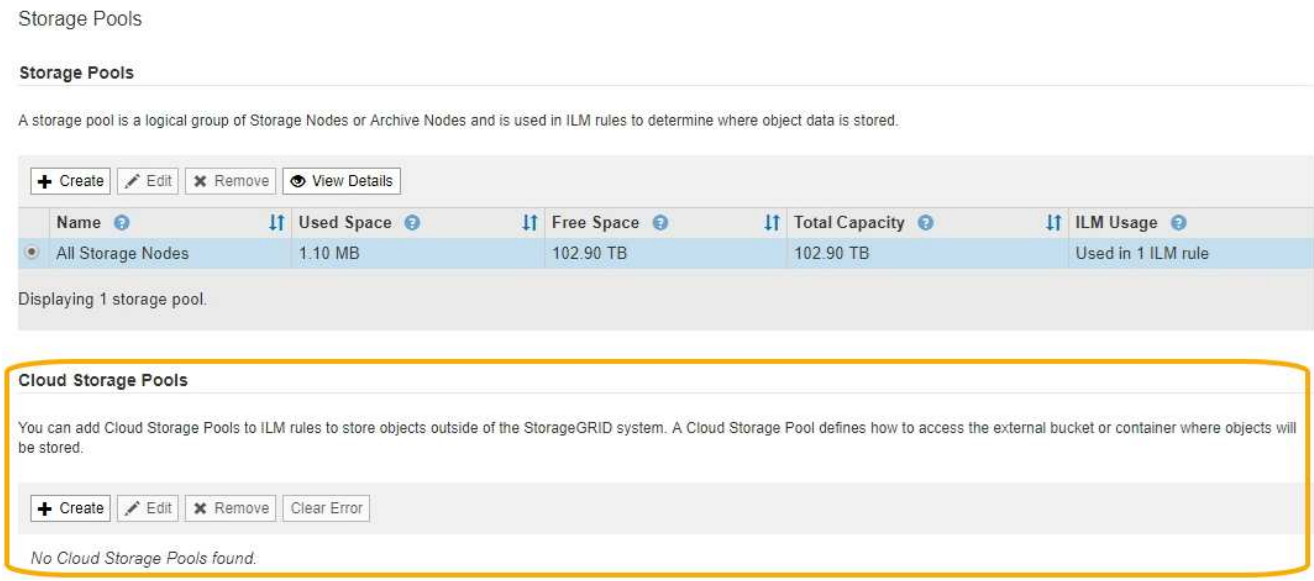
About this task

A Cloud Storage Pool specifies a single external S3 bucket or Azure Blob storage container. StorageGRID validates the Cloud Storage Pool as soon as you save it, so you must ensure that the bucket or container specified in the Cloud Storage Pool exists and is reachable.

Steps

1. Select **ILM** > **Storage Pools**.

   The Storage Pools page appears. This page includes two sections: Storage Pools and Cloud Storage Pools.



2. In the Cloud Storage Pools section of the page, click **Create**.

   The Create Cloud Storage Pool dialog box appears.

3. Enter the following information:

Field	Description
Display Name	A name that briefly describes the Cloud Storage Pool and its purpose. Use a name that will be easy to identify when you configure ILM rules.
Provider Type	Which cloud provider you will use for this Cloud Storage Pool: • Amazon S3 (select this option for an S3 or C2S S3 Cloud Storage Pool) • Azure Blob Storage **Note:** When you select a Provider Type, the Service Endpoint, Authentication and Server Verification sections appear at the bottom on the page.

Field	Description
Bucket or Container	The name of the external S3 bucket or Azure container that was created for the Cloud Storage Pool. The name you specify here must exactly match the bucket or container's name or Cloud Storage Pool creation will fail. You cannot change this value after the Cloud Storage Pool is saved.

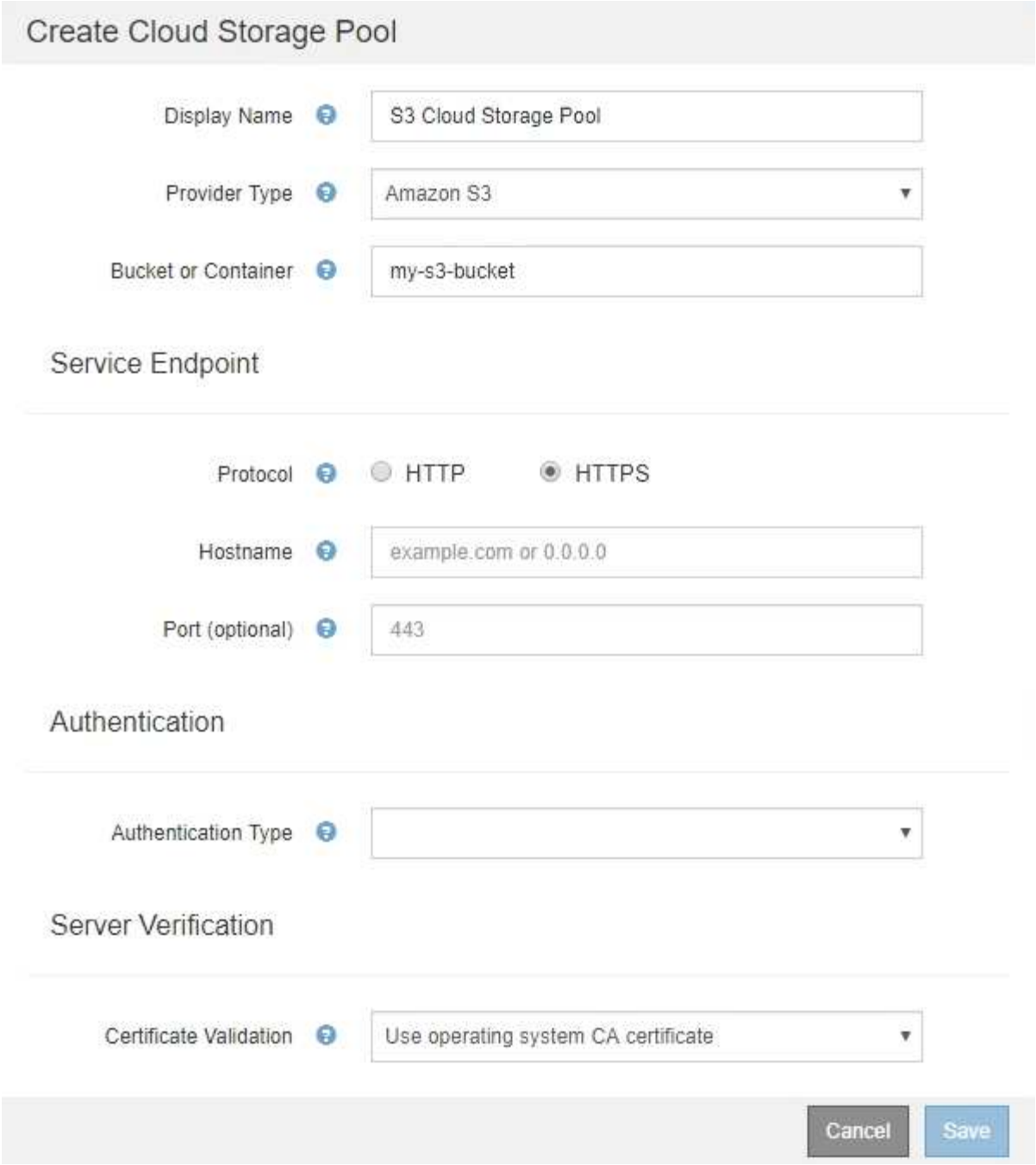
4. Complete the Service Endpoint, Authentication and Server Verification sections of the page, based on the selected provider type.
   - S3: Specifying authentication details for a Cloud Storage Pool
   - C2S S3: Specifying authentication details for a Cloud Storage Pool
   - Azure: Specifying authentication details for a Cloud Storage Pool

S3: Specifying authentication details for a Cloud Storage Pool

When you create a Cloud Storage Pool for S3, you must select the type of authentication that is required for the Cloud Storage Pool endpoint. You can specify Anonymous or enter an Access Key ID and Secret Access Key.

What you'll need

- You must have entered the basic information for the Cloud Storage Pool and specified **Amazon S3** as the provider type.



- If you are using access key authentication, you must know the Access Key ID and Secret Access Key for the external S3 bucket.

Steps

1. In the **Service Endpoint** section, provide the following information:
   a. Select which protocol to use when connecting to the Cloud Storage Pool.

      The default protocol is HTTPS.

   b. Enter the server hostname or IP address of the Cloud Storage Pool.

      For example:

      `s3-aws-region.amazonaws.com`

> Do not include the bucket name in this field. You include the bucket name in the **Bucket or Container** field.

c. Optionally, specify the port that should be used when connecting to the Cloud Storage Pool.

   Leave this field blank to use the default port: port 443 for HTTPS or port 80 for HTTP.

2. In the **Authentication** section, select the type of authentication that is required for the Cloud Storage Pool endpoint.

Option	Description
Access Key	An Access Key ID and Secret Access Key are required to access the Cloud Storage Pool bucket.
Anonymous	Everyone has access to the Cloud Storage Pool bucket. An Access Key ID and Secret Access Key are not required.
CAP (C2S Access Portal)	Used for C2S S3 only. Go to C2S S3: Specifying authentication details for a Cloud Storage Pool.

3. If you selected Access Key, enter the following information:

Option	Description
Access Key ID	The Access Key ID for the account that owns the external bucket.
Secret Access Key	The associated Secret Access Key.

4. In the Server Verification section, select which method should be used to validate the certificate for TLS connections to the Cloud Storage Pool:

Option	Description
Use operating system CA certificate	Use the default CA certificates installed on the operating system to secure connections.
Use custom CA certificate	Use a custom CA certificate. Click **Select New**, and upload the PEM-encoded CA certificate.
Do not verify certificate	The certificate used for the TLS connection is not verified.

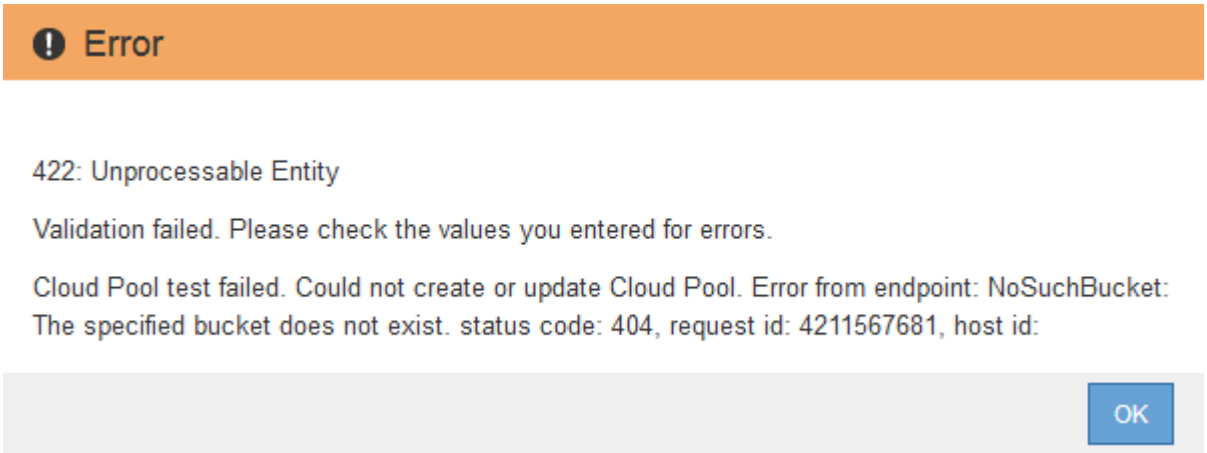
5. Click **Save**.

When you save a Cloud Storage Pool, StorageGRID does the following:

- Validates that the bucket and the service endpoint exist and that they can be reached using the credentials that you specified.
- Writes a marker file to the bucket to identify the bucket as a Cloud Storage Pool. Never remove this file,

which is named `x-ntap-sgws-cloud-pool-uuid`.

If Cloud Storage Pool validation fails, you receive an error message that explains why validation failed. For example, an error might be reported if there is a certificate error or if the bucket you specified does not already exist.



See the instructions for troubleshooting Cloud Storage Pools, resolve the issue, and then try saving the Cloud Storage Pool again.

Related information

Troubleshooting Cloud Storage Pools

C2S S3: Specifying authentication details for a Cloud Storage Pool

To use the Commercial Cloud Services (C2S) S3 service as a Cloud Storage Pool, you must configure C2S Access Portal (CAP) as the authentication type, so that StorageGRID can request temporary credentials to access the S3 bucket in your C2S account.

What you'll need

- You must have entered the basic information for an Amazon S3 Cloud Storage Pool, including the service endpoint.
- You must know the complete URL that StorageGRID will use to obtain temporary credentials from the CAP server, including all the required and optional API parameters assigned to your C2S account.
- You must have a server CA certificate issued by an appropriate Government Certificate Authority (CA). StorageGRID uses this certificate to verify the identity of the CAP server. The server CA certificate must use PEM encoding.
- You must have a client certificate issued by an appropriate Government Certificate Authority (CA). StorageGRID uses this certificate to identity itself to the CAP server. The client certificate must use PEM encoding and must have been granted access to your C2S account.
- You must have a PEM-encoded private key for the client certificate.
- If the private key for the client certificate is encrypted, you must have the passphrase for decrypting it.

Steps

1. In the **Authentication** section, select **CAP (C2S Access Portal)** from the **Authentication Type** drop-down.

   The CAP C2S authentication fields appear.

Create Cloud Storage Pool

Display Name: S3 Cloud Storage Pool

Provider Type: Amazon S3

Bucket or Container: my-s3-bucket

Service Endpoint

Protocol: HTTP HTTPS

Hostname: s3-aws-region.amazonaws.com

Port (optional): 443

Authentication

Authentication Type: CAP (C2S Access Portal)

Temporary Credentials URL: https://example.com/CAP/api/v1/credentials?agency=my

Server CA Certificate: Select New

Client Certificate: Select New

Client Private Key: Select New

Client Private Key Passphrase (optional):

Server Verification

Certificate Validation: Use operating system CA certificate

Cancel Save

2. Provide the following information:
   a. For **Temporary Credentials URL**, enter the complete URL that StorageGRID will use to obtain temporary credentials from the CAP server, including all the required and optional API parameters assigned to your C2S account.
   b. For **Server CA Certificate**, click **Select New**, and upload the PEM-encoded CA certificate that StorageGRID will use to verify the CAP server.
   c. For **Client Certificate**, click **Select New**, and upload the PEM-encoded certificate that StorageGRID will use to identify itself to the CAP server.
   d. For **Client Private Key**, click **Select New**, and upload the PEM-encoded private key for the client certificate.

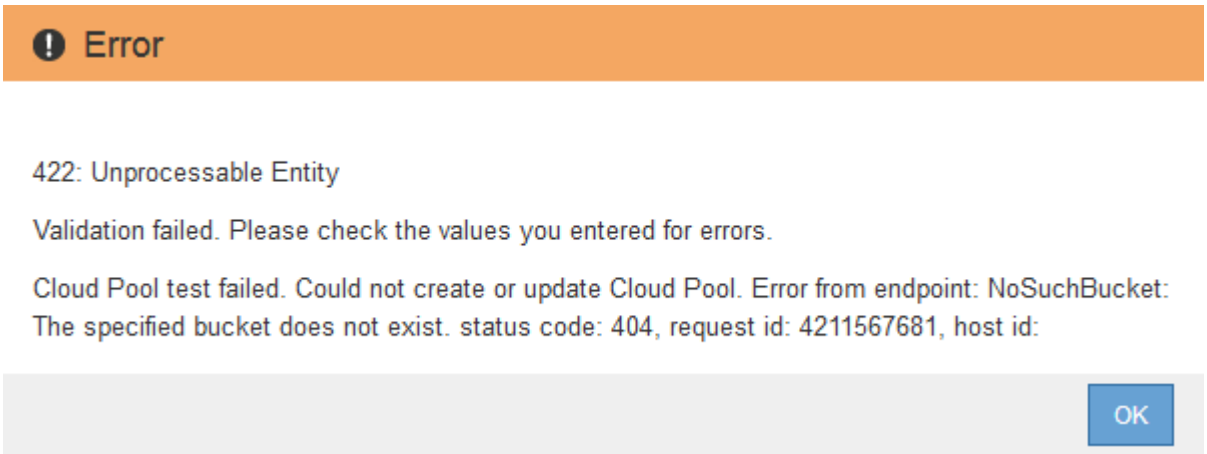
      If the private key is encrypted, the traditional format must be used. (PKCS #8 encrypted format is not supported.)

   e. If the client private key is encrypted, enter the passphrase for decrypting the client private key. Otherwise, leave the **Client Private Key Passphrase** field blank.
3. In the Server Verification section, provide the following information:
   a. For **Certificate Validation**, select **Use custom CA certificate**.
   b. Click **Select New**, and upload the PEM-encoded CA certificate.
4. Click **Save**.

When you save a Cloud Storage Pool, StorageGRID does the following:

- Validates that the bucket and the service endpoint exist and that they can be reached using the credentials that you specified.
- Writes a marker file to the bucket to identify the bucket as a Cloud Storage Pool. Never remove this file, which is named `x-ntap-sgws-cloud-pool-uuid`.

If Cloud Storage Pool validation fails, you receive an error message that explains why validation failed. For example, an error might be reported if there is a certificate error or if the bucket you specified does not already exist.



See the instructions for troubleshooting Cloud Storage Pools, resolve the issue, and then try saving the Cloud Storage Pool again.

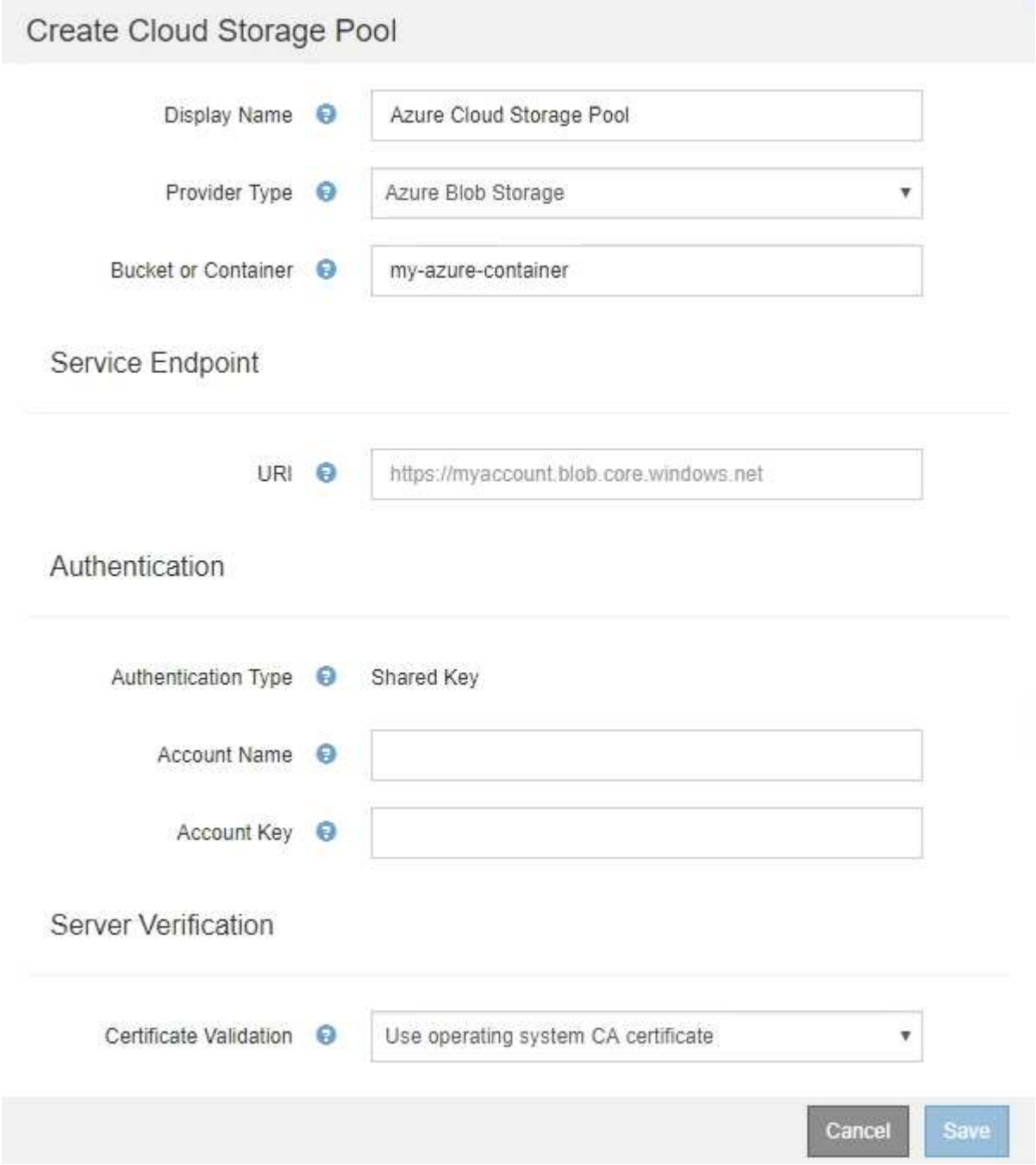
Related information

Azure: Specifying authentication details for a Cloud Storage Pool

When you create a Cloud Storage Pool for Azure Blob storage, you must specify an account name and account key for the external container that StorageGRID will use to store objects.

What you'll need

- You must have entered the basic information for the Cloud Storage Pool and specified **Azure Blob Storage** as the provider type. **Shared Key** appears in the **Authentication Type** field.



- You must know the Uniform Resource Identifier (URI) used to access the Blob storage container used for the Cloud Storage Pool.

- You must know the name of the storage account and the secret key. You can use the Azure portal to find these values.

Steps

1. In the **Service Endpoint** section, enter the Uniform Resource Identifier (URI) used to access the Blob storage container used for the Cloud Storage Pool.

   Specify the URI in one of the following formats:

   - `https://host:port`
   - `http://host:port`

   If you do not specify a port, by default port 443 is used for HTTPS URIs and port 80 is used for HTTP URIs.

   Example URI for Azure Blob storage container: `https://myaccount.blob.core.windows.net`

2. In the **Authentication** section, provide the following information:

   a. For **Account Name**, enter the name of the Blob storage account that owns the external service container.

   b. For **Account Key**, enter the secret key for the Blob storage account.

   For Azure endpoints, you must use Shared Key authentication.

3. In the **Server Verification** section, select which method should be used to validate the certificate for TLS connections to the Cloud Storage Pool:

Option	Description
Use operating system CA certificate	Use the default CA certificates installed on the operating system to secure connections.
Use custom CA certificate	Use a custom CA certificate. Click **Select New**, and upload the PEM-encoded certificate.
Do not verify certificate	The certificate used for the TLS connection is not verified.

4. Click **Save**.

When you save a Cloud Storage Pool, StorageGRID does the following:

- Validates that the container and the URI exist and that they can be reached using the credentials that you specified.
- Writes a marker file to the container to identify it as a Cloud Storage Pool. Never remove this file, which is named `x-ntap-sgws-cloud-pool-uuid`.

If Cloud Storage Pool validation fails, you receive an error message that explains why validation failed. For example, an error might be reported if there is a certificate error or if the container you specified does not already exist.

See the instructions for troubleshooting Cloud Storage Pools, resolve the issue, and then try saving the Cloud Storage Pool again.

Related information

Troubleshooting Cloud Storage Pools

Editing a Cloud Storage Pool

You can edit a Cloud Storage Pool to change its name, service endpoint, or other details; however, you cannot change the S3 bucket or Azure container for a Cloud Storage Pool.

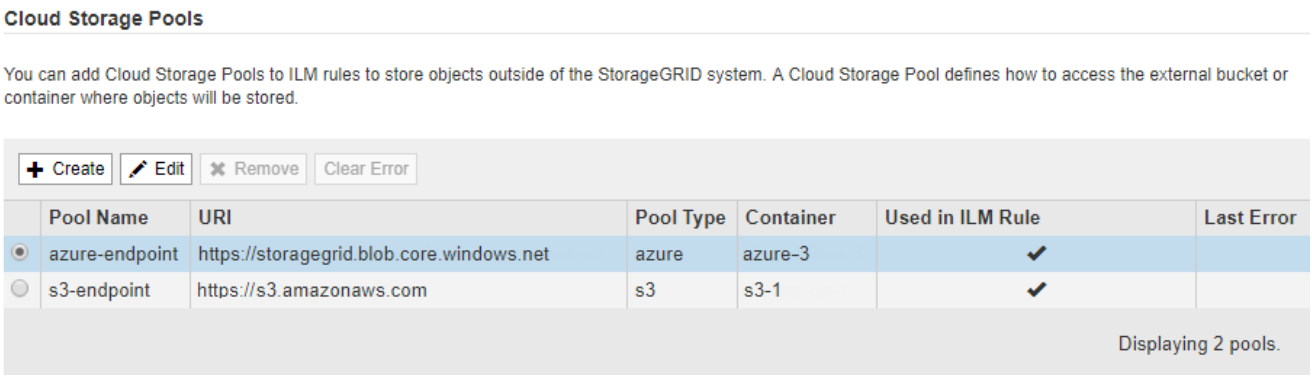
What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have reviewed the guidelines for configuring Cloud Storage Pools.

Steps

1. Select **ILM** > **Storage Pools**.

   The Storage Pools page appears. The Cloud Storage Pools table lists the existing Cloud Storage Pools.



2. Select the radio button for the Cloud Storage Pool you want to edit.
3. Click **Edit**.
4. As required, change the display name, service endpoint, authentication credentials, or certificate validation method.

   > You cannot change the provider type or the S3 bucket or Azure container for a Cloud Storage Pool.

   If you previously uploaded a server or client certificate, you can select **View Current** to review the certificate that is currently in use.

5. Click **Save**.

   When you save a Cloud Storage Pool, StorageGRID validates that the bucket or container and the service endpoint exist, and that they can be reached using the credentials that you specified.

   If Cloud Storage Pool validation fails, an error message is displayed. For example, an error might be reported if there is a certificate error.

See the instructions for troubleshooting Cloud Storage Pools, resolve the issue, and then try saving the Cloud Storage Pool again.

Related information

Considerations for Cloud Storage Pools

Troubleshooting Cloud Storage Pools

Removing a Cloud Storage Pool

You can remove a Cloud Storage Pool that is not used in an ILM rule and that does not contain object data.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You have confirmed that the S3 bucket or Azure container does not contain any objects. An error occurs if you attempt to remove a Cloud Storage Pool if it contains objects. See "Troubleshooting Cloud Storage Pools."

> When you create a Cloud Storage Pool, StorageGRID writes a marker file to the bucket or container to identify it as a Cloud Storage Pool. Do not remove this file, which is named `x-ntap-sgws-cloud-pool-uuid`.

- You have already removed any ILM rules that might have used the pool.

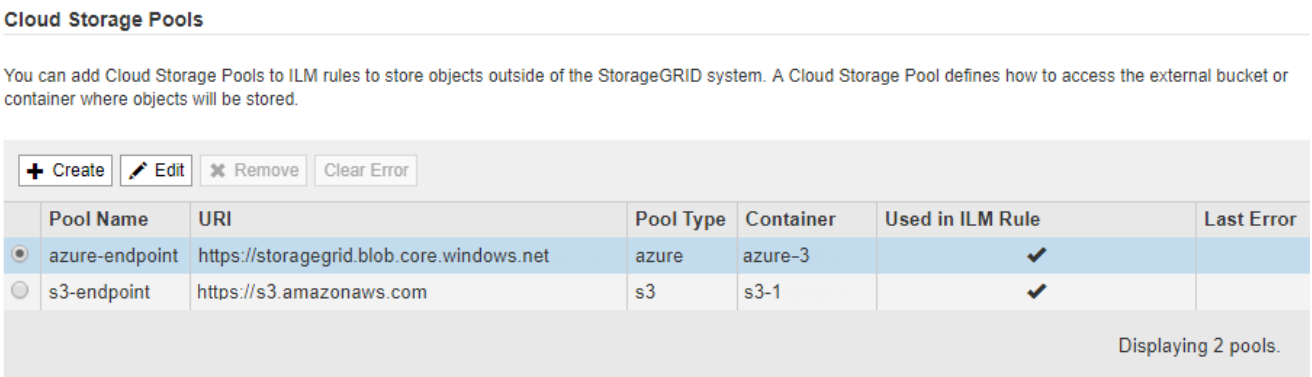
Steps

1. Select **ILM** > **Storage Pools**.

   The Storage Pools page appears.

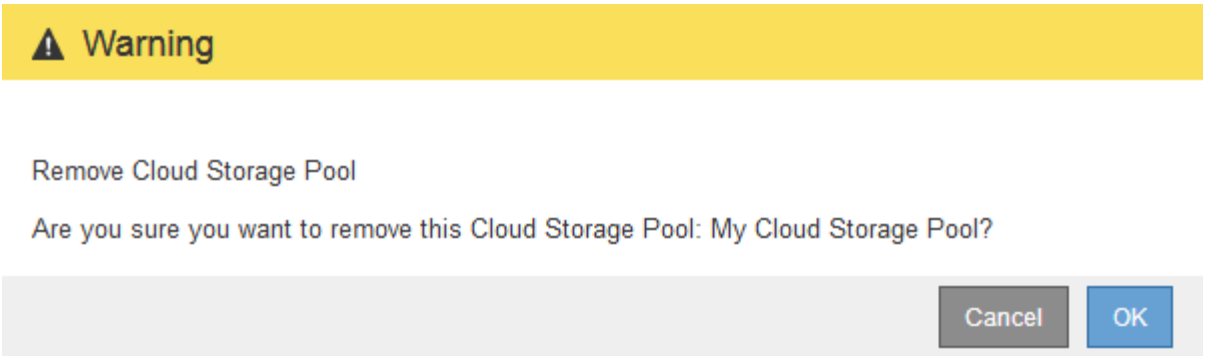
2. Select the radio button for a Cloud Storage Pool that is not currently used in an ILM rule.

   You cannot remove a Cloud Storage Pool if it is used in an ILM rule. The **Remove** button is disabled.



3. Click **Remove**.

   A confirmation warning is displayed.



4. Click **OK**.

   The Cloud Storage Pool is removed.

Related information

Troubleshooting Cloud Storage Pools

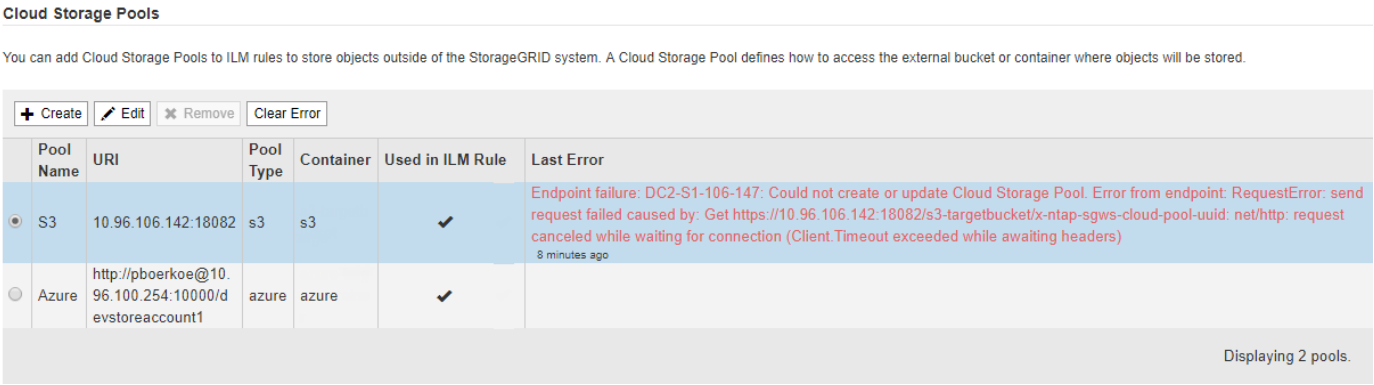
Troubleshooting Cloud Storage Pools

If you encounter errors when creating, editing, or deleting a Cloud Storage Pool, use these troubleshooting steps to help resolve the issue.

Determining if an error has occurred

StorageGRID performs a simple health check on every Cloud Storage Pool once a minute to ensure that the Cloud Storage Pool can be accessed and that it is functioning correctly. If the health check detects an issue, a message is shown in the Last Error column of the Cloud Storage Pools table on the Storage Pools page.

The table shows the most recent error detected for each Cloud Storage Pool and indicates how long ago the error occurred.

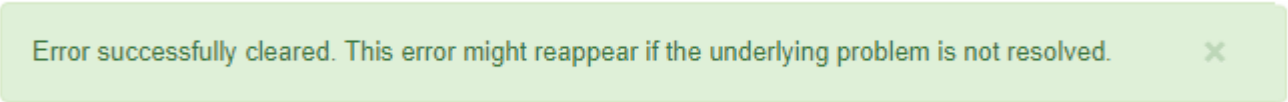


In addition, a **Cloud Storage Pool connectivity error** alert is triggered if the health check detects that one or more new Cloud Storage Pool errors have occurred within the past 5 minutes. If you receive an email notification for this alert, go to the Storage Pool page (select **ILM** > **Storage Pools**), review the error messages in the Last Error column, and refer to the troubleshooting guidelines below.

Checking if an error has been resolved

After resolving any underlying issues, you can determine if the error has been resolved. From the Cloud Storage Pool page, select the radio button for the endpoint, and click **Clear Error**. A confirmation message

indicates that StorageGRID has cleared the error for the Cloud Storage Pool.



If the underlying problem has been resolved, the error message is no longer displayed. However, if the underlying problem has not been fixed (or if a different error is encountered), the error message will be shown in the Last Error column within a few minutes.

Error: This Cloud Storage Pool contains unexpected content

You might encounter this error when you try to create, edit, or delete a Cloud Storage Pool. This error occurs if the bucket or container includes the `x-ntap-sgws-cloud-pool-uuid` marker file, but that file does not have the expected UUID.

Typically, you will only see this error if you are creating a new Cloud Storage Pool and another instance of StorageGRID is already using the same Cloud Storage Pool.

Try these steps to correct the issue:

- Check to make sure that no one in your organization is also using this Cloud Storage Pool.
- Delete the `x-ntap-sgws-cloud-pool-uuid` file and try configuring the Cloud Storage Pool again.

Error: Could not create or update Cloud Storage Pool. Error from endpoint

You might encounter this error when you try to create or edit a Cloud Storage Pool. This error indicates that some kind of connectivity or configuration issue is preventing StorageGRID from writing to the Cloud Storage Pool.

To correct the issue, review the error message from the endpoint.

- If the error message contains `Get *url*: EOF`, check that the service endpoint used for the Cloud Storage Pool does not use the HTTP protocol for a container or bucket that requires HTTPS.
- If the error message contains `Get *url*: net/http: request canceled while waiting for connection`, verify that the network configuration allows Storage Nodes to access the service endpoint used for the Cloud Storage Pool.
- For all other endpoint error messages, try one or more of the following:
  - Create an external container or bucket with the same name you entered for the Cloud Storage Pool, and try to save the new Cloud Storage Pool again.
  - Correct the container or bucket name you specified for the Cloud Storage Pool, and try to save the new Cloud Storage Pool again.

Error: Failed to parse CA certificate

You might encounter this error when you try to create or edit a Cloud Storage Pool. The error occurs if StorageGRID could not parse the certificate you entered when configuring the Cloud Storage Pool.

To correct the issue, check the CA certificate you provided for issues.

Error: A Cloud Storage Pool with this ID was not found

You might encounter this error when you try to edit or delete a Cloud Storage Pool. This error occurs if the endpoint returns a 404 response, which can mean either of the following:

- The credentials used for the Cloud Storage Pool do not have read permission for the bucket.
- The bucket used for the Cloud Storage Pool does not include the `x-ntap-sgws-cloud-pool-uuid` marker file.

Try one or more of these steps to correct the issue:

- Check that the user associated with the configured Access Key has the requisite permissions.
- Edit the Cloud Storage Pool with credentials that have the requisite permissions.
- If the permissions are correct, contact support.

Error: Could not check the content of the Cloud Storage Pool. Error from endpoint

You might encounter this error when you try to delete a Cloud Storage Pool. This error indicates that some kind of connectivity or configuration issue is preventing StorageGRID from reading the contents of the Cloud Storage Pool bucket.

To correct the issue, review the error message from the endpoint.

Error: Objects have already been placed in this bucket

You might encounter this error when you try to delete a Cloud Storage Pool. You cannot delete a Cloud Storage Pool if it contains data that was moved there by ILM, data that was in the bucket before you configured the Cloud Storage Pool, or data that was put in the bucket by some other source after the Cloud Storage Pool was created.

Try one or more of these steps to correct the issue:

- Follow the instructions for moving objects back to StorageGRID in “Lifecycle of a Cloud Storage Pool object.”
- If you are certain the remaining objects were not placed in the Cloud Storage Pool by ILM, manually delete the objects from the bucket.

> Never manually delete objects from a Cloud Storage Pool that might have been placed there by ILM. If you later attempt to access a manually deleted object from StorageGRID, the deleted object will not be found.

Error: Proxy encountered an external error while trying to reach the Cloud Storage Pool

You might encounter this error if you have configured a non-transparent Storage proxy between Storage Nodes and the external S3 endpoint used for the Cloud Storage Pool. This error occurs if the external proxy server cannot reach the Cloud Storage Pool endpoint. For example, the DNS server might not be able to resolve the hostname or there might be an external networking issue.

Try one or more of these steps to correct the issue:

- Check the settings for the Cloud Storage Pool (**ILM** > **Storage Pools**).

- Check the networking configuration of the Storage proxy server.

Related information

Lifecycle of a Cloud Storage Pool object

Configuring Erasure Coding profiles

You configure Erasure Coding profiles by associating a storage pool with an erasure-coding scheme, such as 6+3. Then, when you configure the placement instructions for an ILM rule, you can select the Erasure Coding profile. If an object matches the rule, data and parity fragments are created and distributed to the storage locations in the storage pool according to the erasure-coding scheme.

- Creating an Erasure Coding profile
- Renaming an Erasure Coding profile
- Deactivating an Erasure Coding profile

Creating an Erasure Coding profile

To create an Erasure Coding profile, you associate a storage pool containing Storage Nodes with an erasure-coding scheme. This association determines the number of data and parity fragments created and where the system distributes these fragments.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have created a storage pool that includes exactly one site or a storage pool that includes three or more sites. No erasure-coding schemes are available for a storage pool that has only two sites.

About this task

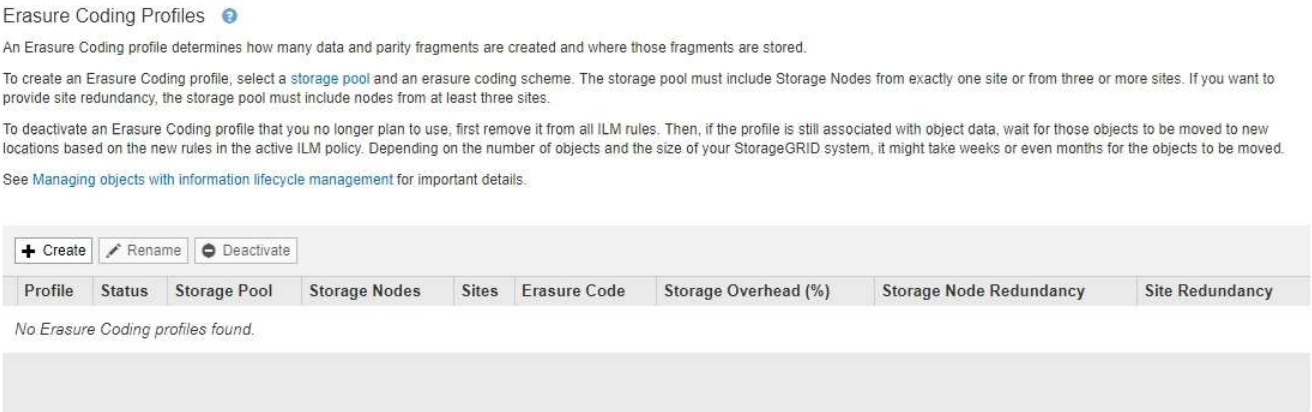
The storage pools used in Erasure Coding profiles must include exactly one site or three or more sites. If you want to provide site redundancy, the storage pool must have at least three sites.

> You must select a storage pool that contains Storage Nodes. You cannot use Archive Nodes for erasure-coded data.

Steps

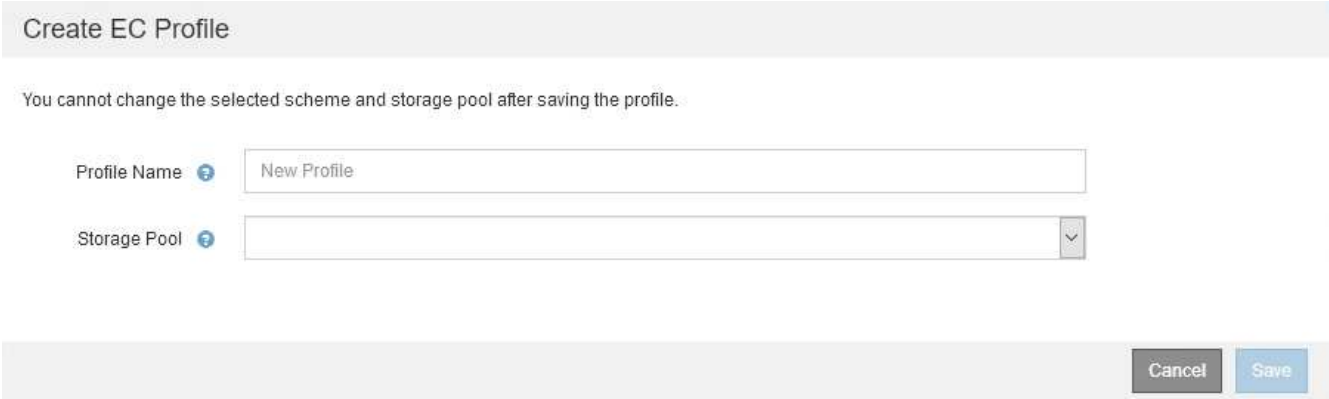
1. Select **ILM** > **Erasure Coding**.

   The Erasure Coding Profiles page appears.



2. Click **Create**.

   The Create EC Profile dialog box appears.



3. Enter a unique name for the Erasure Coding profile.

   Erasure Coding profile names must be unique. A validation error occurs if you use the name of an existing profile, even if that profile has been deactivated.

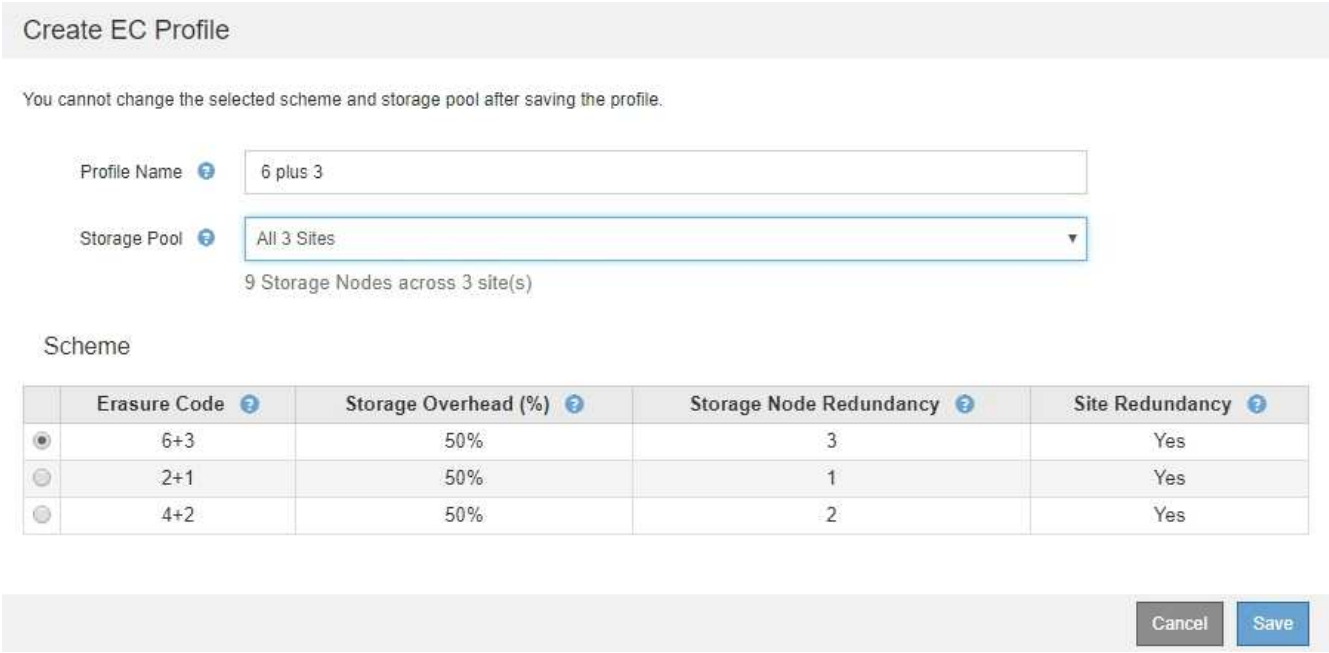
   > The Erasure Coding profile name is appended to the storage pool name in the placement instruction for an ILM rule.

4. Select the storage pool you created for this Erasure Coding profile.

   > If your grid currently includes only one site, you are prevented from using the default storage pool, All Storage Nodes, or any storage pool that includes the default site, All Sites. This behavior prevents the Erasure Coding profile from becoming invalid if a second site is added.

   > If a storage pool includes exactly two sites, you cannot use that storage pool for erasure coding. No erasure-coding schemes are available for a storage pool that has two sites.

When you select a storage pool, the list of available erasure-coding schemes is shown, based on the number of Storage Nodes and sites in the pool.



The following information is listed for each available erasure-coding scheme:

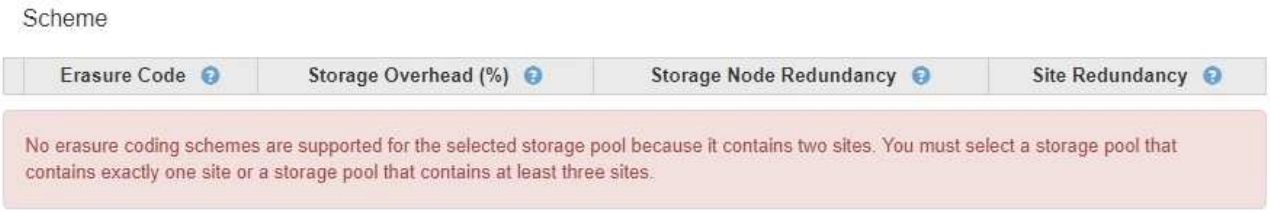
- **Erasure Code**: The name of the erasure-coding scheme in the following format: data fragments + parity fragments.
- **Storage Overhead (%)**: The additional storage required for parity fragments relative to the object's data size. Storage Overhead = Total number of parity fragments / Total number of data fragments.
- **Storage Node Redundancy**: The number of Storage Nodes that can be lost while still maintaining the ability to retrieve object data.
- **Site Redundancy**: Whether the selected erasure code allows the object data to be retrieved if a site is lost.

  To support site redundancy, the selected storage pool must include multiple sites, each with enough Storage Nodes to allow any site to be lost. For example, to support site redundancy using a 6+3 erasure-coding scheme, the selected storage pool must include at least three sites with at least three Storage Nodes at each site.

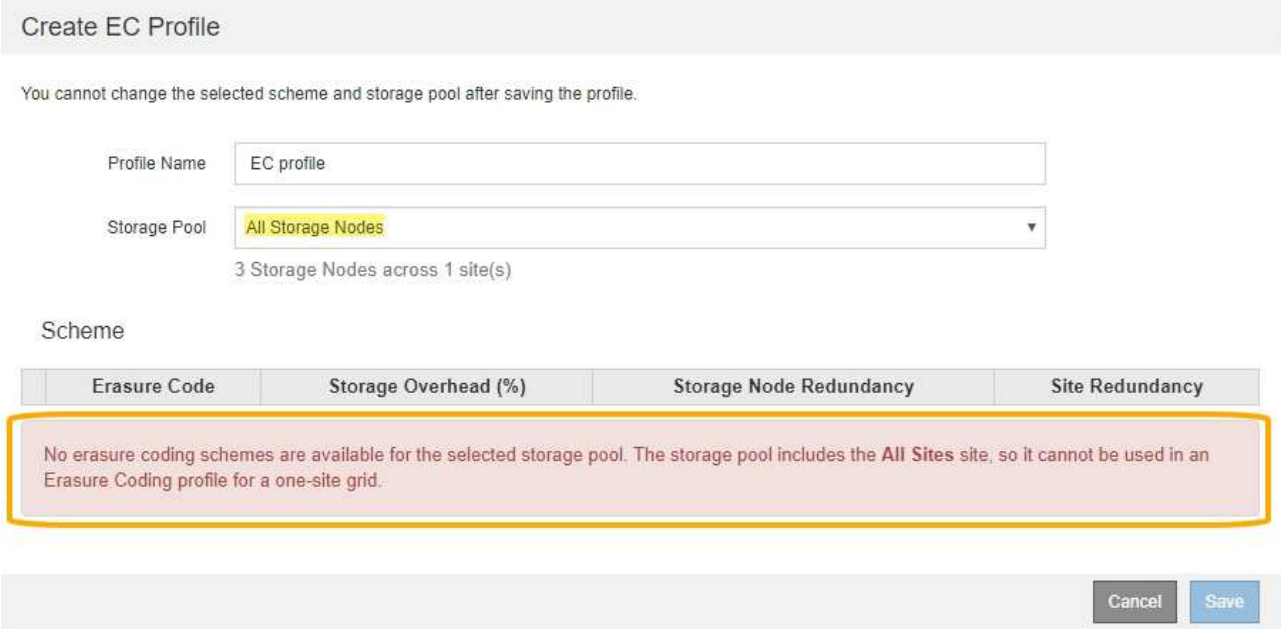
Messages are displayed in these cases:

- The storage pool you selected does not provide site redundancy. The following message is expected when the selected storage pool includes only one site. You can use this Erasure Coding profile in ILM rules to protect against node failures.

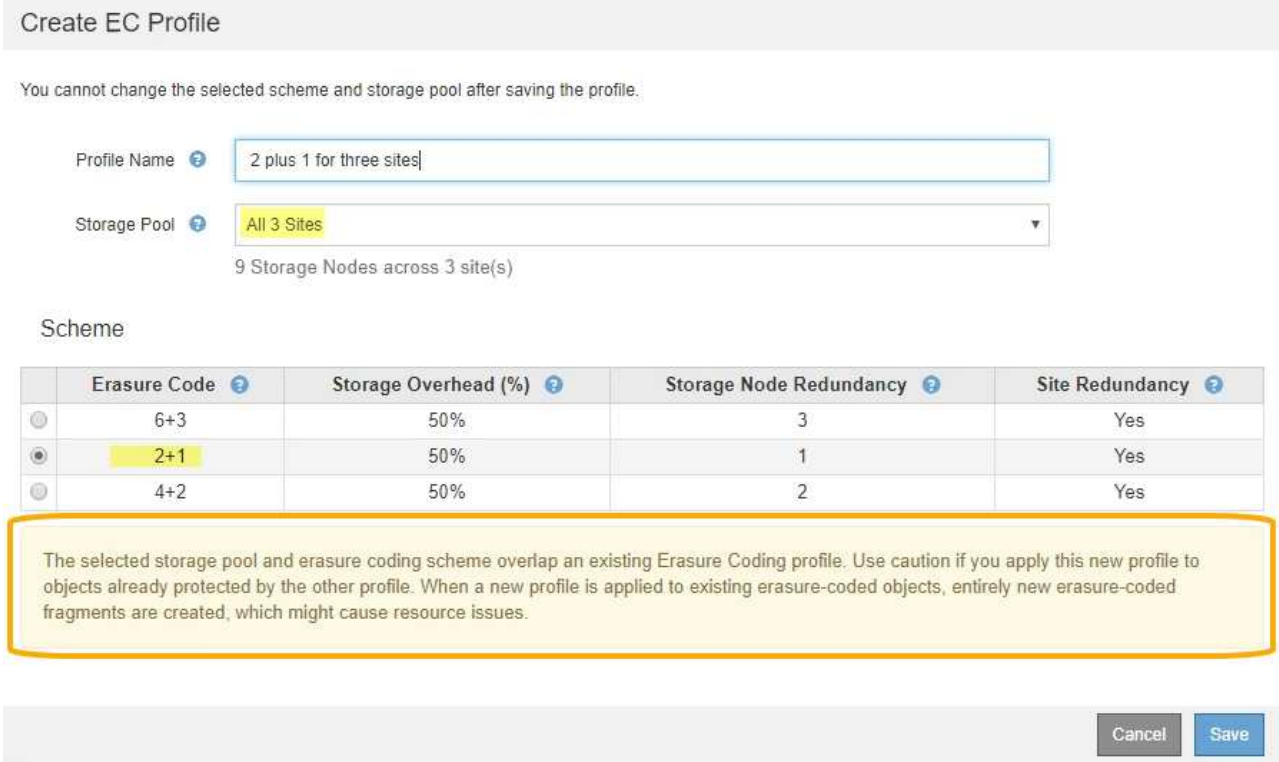
- The storage pool you selected does not satisfy the requirements for any erasure-coding scheme. For example, the following message is expected when the selected storage pool includes exactly two sites. If you want to use erasure coding to protect object data, you must select a storage pool with exactly one site or a storage pool with three or more sites.



- Your grid includes only one site and you selected the default storage pool, All Storage Nodes, or any storage pool that includes the default site, All Sites.



- The erasure-coding scheme and storage pool you selected overlap with another Erasure Coding profile.



In this example, a warning message appears because another Erasure Coding profile is using the 2+1 scheme and the storage pool for the other profile also uses one of the sites in the All 3 Sites storage pool.

While you are not prevented from creating this new profile, you must be very careful when you start using it in the ILM policy. If this new profile is applied to existing erasure-coded objects already protected by the other profile, StorageGRID will create an entirely new set of object fragments. It will not reuse the existing 2+1 fragments. Resource issues might occur when you migrate from one Erasure Coding profile to the other, even though the erasure-coding schemes are the same.

5. If more than one erasure-coding scheme is listed, select the one you want to use.

   When deciding which erasure-coding scheme to use, you should balance fault tolerance (achieved by having more parity segments) against the network traffic requirements for repairs (more fragments equals more network traffic). For example, when deciding between a 4+2 scheme and 6+3 scheme, select the 6+3 scheme if additional parity and fault tolerance are required. Select the 4+2 scheme if network resources are constrained to reduce network usage during node repairs.

6. Click **Save**.

Renaming an Erasure Coding profile

You might want to rename an Erasure Coding profile to make it more obvious what the profile does.

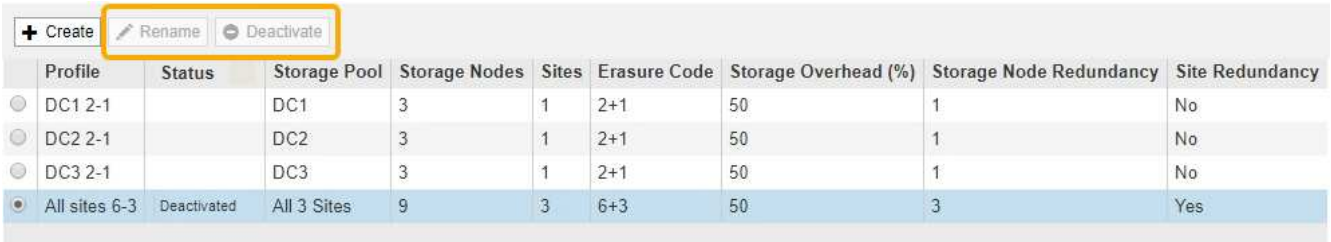
What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.

Steps

1. Select **ILM** > **Erasure Coding**.

   The Erasure Coding Profiles page appears. The **Rename** and **Deactivate** buttons are both disabled.



Profile	Status	Storage Pool	Storage Nodes	Sites	Erasure Code	Storage Overhead (%)	Storage Node Redundancy	Site Redundancy
DC1 2-1		DC1	3	1	2+1	50	1	No
DC2 2-1		DC2	3	1	2+1	50	1	No
DC3 2-1		DC3	3	1	2+1	50	1	No
All sites 6-3	Deactivated	All 3 Sites	9	3	6+3	50	3	Yes

2. Select the profile you want to rename.

   The **Rename** and **Deactivate** buttons become enabled.

3. Click **Rename**.

   The Rename EC Profile dialog box appears.

4. Enter a unique name for the Erasure Coding profile.

   The Erasure Coding profile name is appended to the storage pool name in the placement instruction for an ILM rule.

> Erasure Coding profile names must be unique. A validation error occurs if you use the name of an existing profile, even if that profile has been deactivated.

5. Click **Save**.

Deactivating an Erasure Coding profile

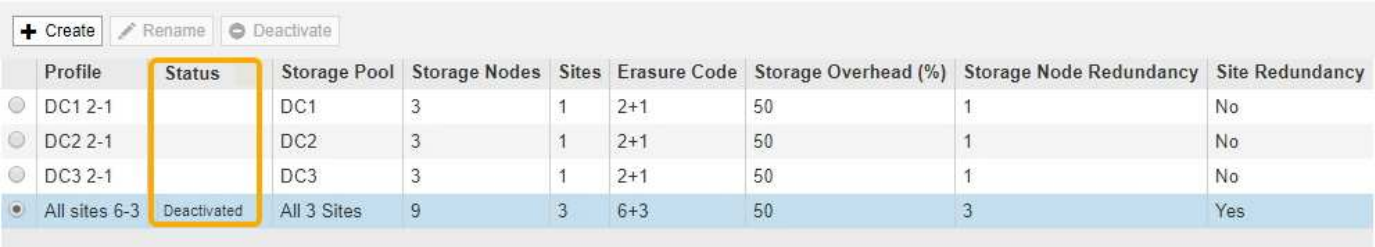
You can deactivate an Erasure Coding profile if you no longer plan to use it and if the profile is not currently used in any ILM rules.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.
- You must have confirmed that no erasure coded data repair operations or decommission procedures are in process. An error message is returned if you attempt to deactivate an Erasure Coding profile while either of these operations are in progress.

About this task

When you deactivate an Erasure Coding profile, the profile still appears on the Erasure Coding Profiles page, but its status is **Deactivated**.



Profile	Status	Storage Pool	Storage Nodes	Sites	Erasure Code	Storage Overhead (%)	Storage Node Redundancy	Site Redundancy
DC1 2-1		DC1	3	1	2+1	50	1	No
DC2 2-1		DC2	3	1	2+1	50	1	No
DC3 2-1		DC3	3	1	2+1	50	1	No
All sites 6-3	Deactivated	All 3 Sites	9	3	6+3	50	3	Yes

You can no longer use an Erasure Coding profile that has been deactivated. A deactivated profile is not shown when you create the placement instructions for an ILM rule. You cannot reactivate a deactivated profile.

StorageGRID prevents you from deactivating an Erasure Coding profile if either of the following is true:

- The Erasure Coding profile is currently used in an ILM rule.
- The Erasure Coding profile is no longer used in any ILM rules, but object data and parity fragments for the profile still exist.

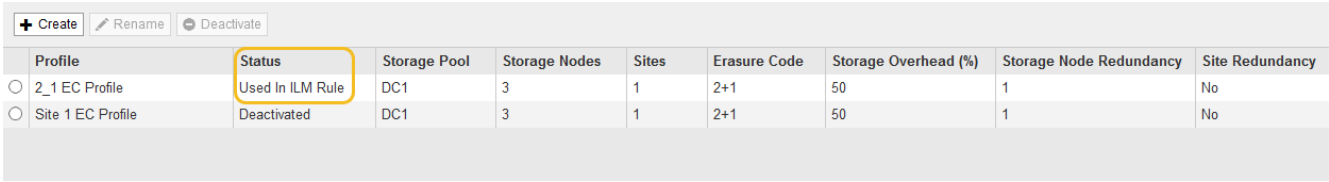
Steps

1. Select **ILM** > **Erasure Coding**.

   The Erasure Coding Profiles page appears. The **Rename** and **Deactivate** buttons are both disabled.

2. Review the **Status** column to confirm that the Erasure Coding profile you want to deactivate is not used in any ILM rules.

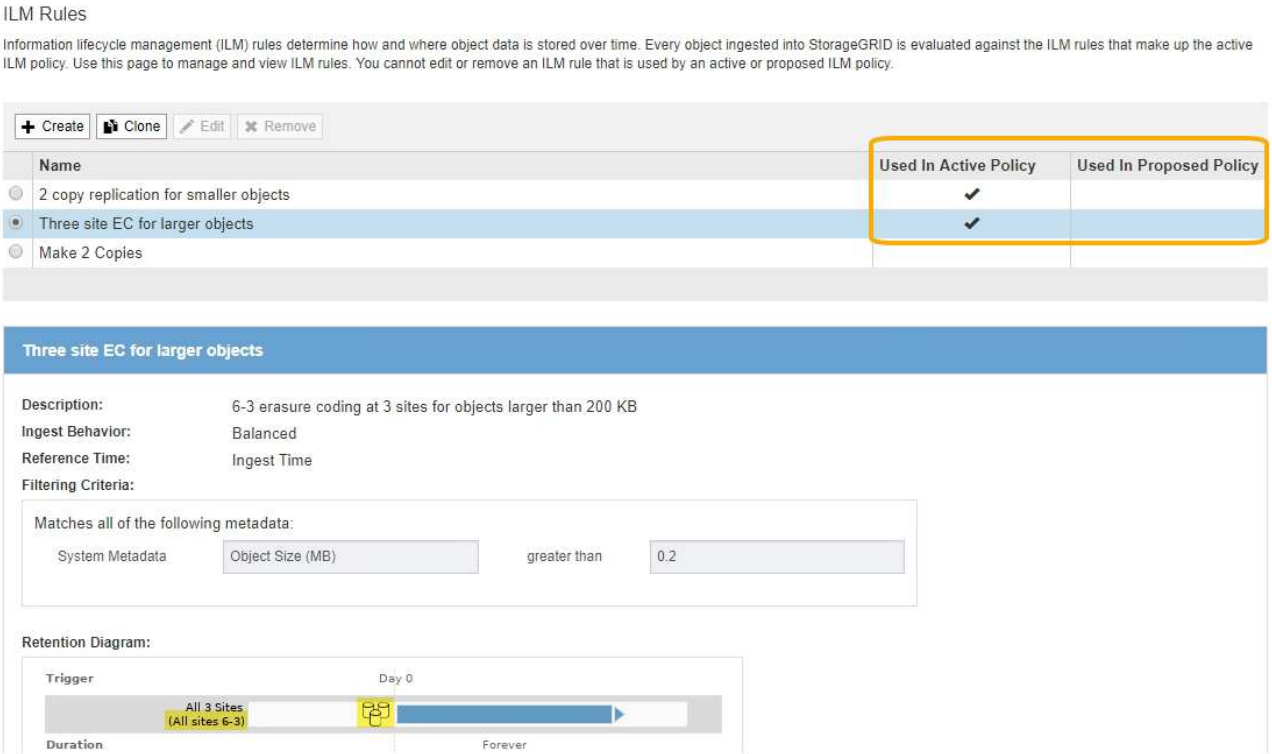
   You cannot deactivate an Erasure Coding profile if it is used in any ILM rule. In the example, the **2_1 EC Profile** is used in at least one ILM rule.



Profile	Status	Storage Pool	Storage Nodes	Sites	Erasure Code	Storage Overhead (%)	Storage Node Redundancy	Site Redundancy
2_1 EC Profile	Used In ILM Rule	DC1	3	1	2+1	50	1	No
Site 1 EC Profile	Deactivated	DC1	3	1	2+1	50	1	No

3. If the profile is used in an ILM rule, follow these steps:
   a. Select **ILM** > **Rules**.
   b. For each rule listed, select the radio button and review the retention diagram to determine if the rule uses the Erasure Coding profile you want to deactivate.

      In the example, the **Three site EC for larger objects** rule uses a storage pool called **All 3 Sites** and the **All sites 6-3** Erasure Coding profile. Erasure Coding profiles are represented by this icon:



c. If the ILM rule uses the Erasure Coding profile you want to deactivate, determine if the rule is used in either the active ILM policy or a proposed policy.

   In the example, the **Three site EC for larger objects** rule is used in the active ILM policy.

d. Complete the additional steps in the table, based on where the Erasure Coding profile is used.

Where has the profile been used?	Additional steps to perform before deactivating the profile	Refer to these additional instructions
Never used in any ILM rule	No additional steps required. Continue with this procedure.	*none*
In an ILM rule that has never been used in any ILM policy	1. Edit or delete all affected ILM rules. If you edit the rule, remove all placements that use the Erasure Coding profile. 2. Continue with this procedure.	Working with ILM rules and ILM policies

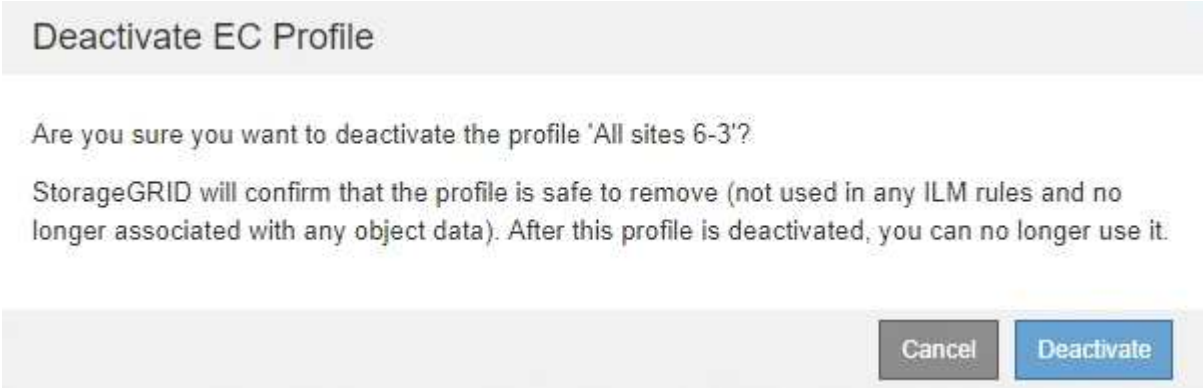
Where has the profile been used?	Additional steps to perform before deactivating the profile	Refer to these additional instructions
In an ILM rule that is currently in the active ILM policy	1. Clone the active policy. 2. Remove the ILM rule that uses the Erasure Coding profile. 3. Add one or more new ILM rules to ensure objects are protected. 4. Save, simulate, and activate the new policy. 5. Wait for the new policy to be applied and for existing objects to be moved to new locations based on the new rules you added. **Note:** Depending on the number of objects and the size of your StorageGRID system, it might take weeks or even months for ILM operations to move the objects to new locations, based on the new ILM rules. While you can safely attempt to deactivate an Erasure Coding profile while it is still associated with data, the deactivation operation will fail. An error message will inform you if the profile is not yet ready to be deactivated. 6. Edit or delete the rule you removed from the policy. If you edit the rule, remove all placements that use the Erasure Coding profile. 7. Continue with this procedure.	• Creating an ILM policy • Working with ILM rules and ILM policies
In an ILM rule that is currently in a proposed ILM policy	1. Edit the proposed policy. 2. Remove the ILM rule that uses the Erasure Coding profile. 3. Add one or more new ILM rules to ensure all objects are protected. 4. Save the proposed policy. 5. Edit or delete the rule you removed from the policy. If you edit the rule, remove all placements that use the Erasure Coding profile. 6. Continue with this procedure.	• Creating an ILM policy • Working with ILM rules and ILM policies

Where has the profile been used?	Additional steps to perform before deactivating the profile	Refer to these additional instructions
In an ILM rule that is in a historical ILM policy	1. Edit or delete the rule. If you edit the rule, remove all placements that use the Erasure Coding profile. (The rule will now appear as a historical rule in the historical policy.) 2. Continue with this procedure.	• Working with ILM rules and ILM policies

e. Refresh the Erasure Coding Profiles page to ensure that the profile is not used in an ILM rule.

4. If the profile is not used in an ILM rule, select the radio button and select **Deactivate**.

   The Deactivate EC Profile dialog box appears.



5. If you are sure you want to deactivate the profile, select **Deactivate**.
   - If StorageGRID is able to deactivate the Erasure Coding profile, its status is **Deactivated**. You can no longer select this profile for any ILM rule.
   - If StorageGRID is not able to deactivate the profile, an error message appears. For example, an error message appears if object data is still associated with this profile. You might need to wait several weeks before trying the deactivation process again.

Configuring regions (optional and S3 only)

ILM rules can filter objects based on the regions where S3 buckets are created, allowing you to store objects from different regions in different storage locations. If you want to use an S3 bucket region as a filter in a rule, you must first create the regions that can be used by the buckets in your system.

What you'll need

- You must be signed in to the Grid Manager using a supported browser.
- You must have specific access permissions.

About this task

When creating an S3 bucket, you can specify that the bucket be created in a specific region. Specifying a region allows the bucket to be geographically close to its users, which can help optimize latency, minimize costs, and address regulatory requirements.

When you create an ILM rule, you might want to use the region associated with an S3 bucket as an advanced filter. For example, you can design a rule that applies only to objects in S3 buckets created in the us-west-2 region. You can then specify that copies of those objects be placed on Storage Nodes at a data center site within that region to optimize latency.

When configuring regions, follow these guidelines:

- By default, all buckets are considered to belong to the us-east-1 region.
- You must create the regions using the Grid Manager before you can specify a non-default region when creating buckets using the Tenant Manager or Tenant Management API or with the LocationConstraint request element for S3 PUT Bucket API requests. An error occurs if a PUT Bucket request uses a region that has not been defined in StorageGRID.
- You must use the exact region name when you create the S3 bucket. Region names are case sensitive and must contain at least 2 and no more than 32 characters. Valid characters are numbers, letters, and hyphens.

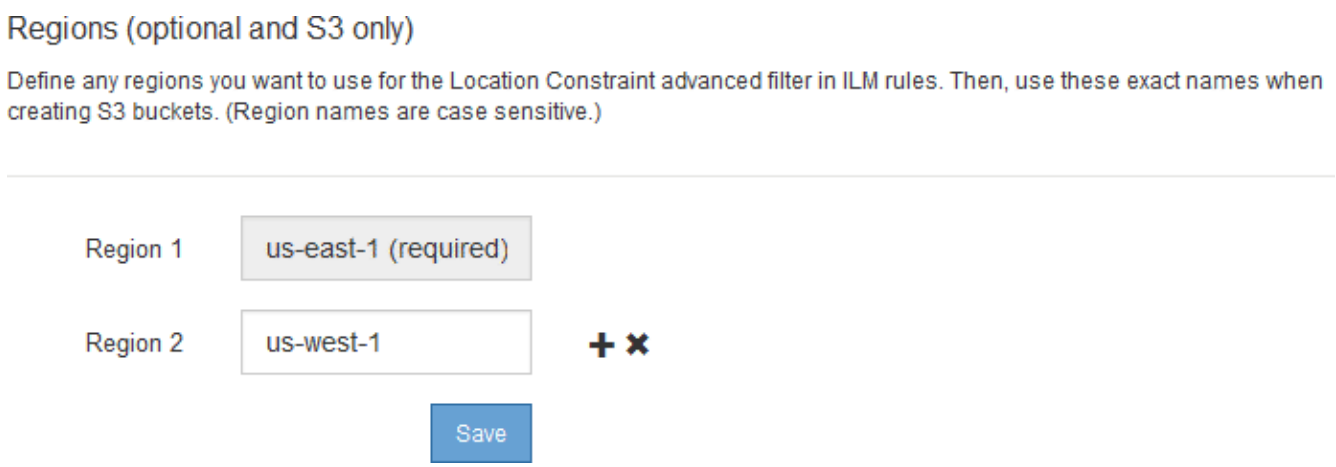
> EU is not considered to be an alias for eu-west-1. If you want to use the EU or eu-west-1 region, you must use the exact name.

- You cannot delete or modify a region if it is currently used within the active ILM policy or the proposed ILM policy.
- If the region used as the advanced filter in an ILM rule is invalid, it is still possible to add that rule to the proposed policy. However, an error occurs if you attempt to save or activate the proposed policy. (An invalid region can result if you use a region as an advanced filter in an ILM rule but you later delete that region, or if you use the Grid Management API to create a rule and specify a region that you have not defined.)
- If you delete a region after using it to create an S3 bucket, you will need to re-add the region if you ever want to use the Location Constraint advanced filter to find objects in that bucket.

Steps

1. Select **ILM** > **Regions**.

   The Regions page appears, with the currently defined regions listed. **Region 1** shows the default region, `us-east-1`, which cannot be modified or removed.

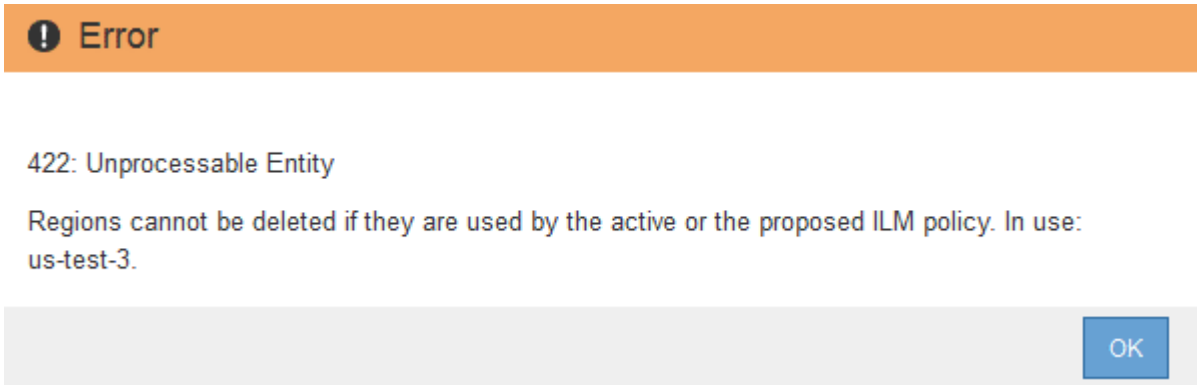
2. To add a region:
   a. Click the insert icon ✚ to the right of the last entry.

b. Enter the name of a region that you want to use when creating S3 buckets.

   You must use this exact region name as the LocationConstraint request element when you create the corresponding S3 bucket.

3. To remove an unused region, click the delete icon ✖.

   An error message appears if you attempt to remove a region that is currently used in the active policy or the proposed policy.



4. When you are done making changes, click **Save**.

   You can now select these regions from the **Location Constraint** list on the Advanced Filtering page of the Create ILM rule wizard.

Related information

Using advanced filters in ILM rules

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